

Institute of Mathematical Sciences, University of Malaya

SIM1001 Basic Mathematics

SIX1013 Fundamentals of Advanced Mathematics

Part 1 Logic

Chapter 1: Fundamentals of Logic

Chapter 2: Set Theory and functions

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Chapter 1 Fundamentals of Logic

1.1 Statements and Compound Statements

Definition 1.1.1: Statement

A *statement* (or *proposition*) is a sentence that is true or false but not both.

Definition 1.1.2: Truth value

The truth or falsity of a statement is called its *truth value*.

Example 1.1.1:

- (a) The following are statements:
- (i) Kuala Lumpur is the capital of Malaysia. (Its truth value is true.)
 - (ii) The sun rises in the West. (Its truth value is false.)
 - (iii) $2 > 5$. (Its truth value is false.)
- (b) The following are not statements:
- (i) The number x is positive.
(When $x = 1$, the sentence is true but when $x = -1$, it is false.)
 - (ii) Is the integer $(2^{10} - 1)$ even?
 - (iii)
 - (iv) Take the book.

In this course, we shall use the letters p , q , r and s to represent statements and these letters are called statement variables.

Connectives

Most mathematical statements are combinations of simpler statement formed through some choice of the words *not*, *and*, *or*, *if*____ *then*____, and *if and only if*. These are called *logical connectives* (or simply *connectives*) and are denoted by following symbols:

- $\sim$ denotes *not* (see **Definition 1.1.4**)
- $\wedge$ denotes *and* (see **Definition 1.1.5**)
- $\vee$ denotes *or* (see **Definition 1.1.6**)
- $\rightarrow$ denotes *if*____, *then*____ (see **Definition 1.3.1**)
- $\leftrightarrow$ denotes *if and only if* (see **Definition 1.3.3**)

Compound statements

Definition 1.1.3: Simple statement and compound statement

A statement represented by a single statement variable (without any connective) is called a *simple* or *primitive* statement.

A statement represented by a some combination of statement variables and connectives is called a *compound statement*.

Example 1.1.2:

- (a) The following are simple statements:
- (i) A cat is an animal.
 - (ii) $3 < 3$.
- (b) The following are compound statements:
- (i) 3 and 5 are odd numbers.
 - (ii) π is not an integer.
 - (iii) $3 \leq 3$.

Let p, q, r be statement variables.

Definition 1.1.4: The negation of p ($\sim p$)

The *negation of p* is “not p ” or “It is not the case that p ” and it is denoted $\sim p$. $\sim p$ has opposite truth value from p .

The truth values for negation of p are summarized in a truth table.

Truth table for $\sim p$

p	$\sim p$
T	F
F	T

T = true, F = false.

Example 1.1.3:

p	$\sim p$
(i) $3 + 3 = 6$.	(i) $3 + 3 \neq 6$. (Or, It is false that $3 + 3 = 6$.)
(ii) 13 is a prime number.	(ii) 13 is not a prime number.
(iii) All integers ending in 3 are prime.	(iii) It is false that all integers ending in 3 are prime.

Definition 1.1.5: The conjunction of p and q ($p \wedge q$)

The *conjunction of p and q* is “ p and q ” and it is denoted $p \wedge q$.

$p \wedge q$ is true when, and only when both p and q are true.

If either p or q is false, then $p \wedge q$ is false.

The truth values for conjunction of p and q are summarized in a truth table.

Truth table for $p \wedge q$

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

But and Neither-Nor

The words “but” and “and” use the same connective $\wedge$. Generally, “but” is used in place of “and” when the part of the sentence that follows in some unexpected way.

The phrase “neither p nor q ” means “not p and not q ” ($\sim p \wedge \sim q$).

$\sim p \wedge \sim q$ does not mean $\sim (p \wedge q)$; and $\sim p \wedge q$ does not mean $\sim (p \wedge q)$.

Example 1.1.4:

Determine the truth value of the following statements.

(i) Kuala Lumpur is in Malaysia and zero is positive.

(ii) The integer 2 is prime but it is even.

(iii) Neither π nor $\sqrt{2}$ is integer.

Definition 1.1.6: The disjunction of p and q ($p \vee q$)

The *disjunction of p and q* is “ p or q ” and it is denoted $p \vee q$.

$p \vee q$ is true when at least one of p or q is true.

$p \vee q$ is false when both p and q are false.

Truth table for $p \vee q$

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Remark:

1. The notation for inequalities involves “and” and “or” statements:

Let a , b and c be particular real numbers.

$a \leq b$ means $a < b$ or $a = b$. (a is less than b or a equals b .)

$a < b < c$ means $a < b$ and $b < c$. (a is less than b and b is less than c .)

2. Whenever we use the word *or* in mathematics, we always mean the inclusive or (p or q or both). The connective “ $\vee$ ” is to denote *inclusive or*.

Example of *inclusive or*: 2 or 5 are integers.

To express the exclusive or the phrase “ p or q but not both” is used.

Example of *exclusive or*: The baby is a boy or a girl, but not both.

Example 1.1.5 :

Determine the truth value of the given statements.

(i) $\sin \pi$ or $\cos \pi$ is zero.

(ii) $3 \leq 3$.

Definition 1.1.7: Proposition and truth table

A **proposition** is a compound statement made up of statement variables (such as p , q and r) and connectives (such as $\sim$, $\wedge$ and $\vee$).

The **truth table** for a given proposition displays the truth values that correspond to the different combinations of truth values for the variables.

Remark:

- (1) Proposition with 2 statement variables p and q

There are 4 rows in the truth table to fill in the truth values. The 4 logically possible combinations of truth values for the pair p , q are TT, TF, FT and FF.

- (2) Proposition with 3 statement variables p , q and r

There are 8 rows in the truth table to fill in the truth values. The 8 logically possible combinations are TTT, TTF, TFT, TFF, FTT, FTF, FFT, and FFF.

Truth table for

p	q	r	
T	T	T	
T	T	F	
T	F	T	
T	F	F	
F	T	T	
F	T	F	
F	F	T	
F	F	F	

- (3) $\wedge$ and $\vee$ are considered coequal (in terms of order of operation).

An expression $p \wedge q \vee r$ is considered meaningless. Parentheses should be added and the expression should be written as $(p \wedge q) \vee r$ or $p \wedge (q \vee r)$. Take note that $(p \wedge q) \vee r$ and $p \wedge (q \vee r)$ are not the same.

Example 1.1.6:

Construct a truth table for the proposition $(p \wedge q) \vee \sim r$.

Steps:

- (1) Set up columns labeled p , q , r , $\sim r$, $(p \wedge q)$ and $(p \wedge q) \vee \sim r$.
- (2) Fill in the p , q , and r columns with all the logically possible combinations of T's and F's.
- (3) Use the truth tables for $\sim$ and $\wedge$ to fill in the $\sim r$ and $(p \wedge q)$ columns with the appropriate truth values.
- (4) Finally, fill in the $(p \wedge q) \vee \sim r$ column.

Truth table for $(p \wedge q) \vee \sim r$

p	q	r	$\sim r$	$(p \wedge q)$	$(p \wedge q) \vee \sim r$
T	T	T	F	T	
T	T	F	T	T	
T	F	T	F	F	
T	F	F	T	F	
F	T	T	F	F	
F	T	F	T	F	
F	F	T	F	F	
F	F	F	T	F	

Tautology and Contradiction

Definition 1.1.8 : Tautology and Contradiction

A proposition is called a **tautology** if it is **true** for any truth values of its variables.
(That is, it contains only “T” in the last column of its truth table.)

A proposition is called a **contradiction** if it is **false** for any truth values of its variables.
(That is, it contains only “F” in the last column of its truth table.)

Remark: Normally, t is used to denote a tautology and
 c is used to denote a contradiction.

Example 1.1.7: Let p , q and r be a statement variable.

(i) **Truth table for $p \vee \sim p$ and $p \wedge \sim p$**

p	$\sim p$	$p \vee \sim p$	$p \wedge \sim p$
T	F	T	F
F	T	T	F

$\therefore p \vee \sim p$ is a tautology and $p \wedge \sim p$ is a contradiction.

(ii) **Truth table for $(p \wedge q) \wedge (\sim p \wedge r)$**

p	q	r	$\sim p$	$p \wedge q$	$\sim p \wedge r$	$(p \wedge q) \wedge (\sim p \wedge r)$
T	T	T	F	T	F	F
T	T	F	F	T	F	F
T	F	T	F	F	F	F
T	F	F	F	F	F	F
F	T	T	T	F	T	F
F	T	F	T	F	F	F
F	F	T	T	F	T	F
F	F	F	T	F	F	F

$\therefore$ The proposition $(p \wedge q) \wedge (\sim p \wedge r)$ is a contradiction.

1.2 Logical Equivalence

Definition 1.2.1: Logical Equivalence ($P \equiv Q$)

Two propositions P and Q are said to be *logically equivalent*, or simply *equivalent* if their truth values agree for all possible combinations of the truth values of the variables.

Notation: $P \equiv Q$.

Example 1.2.1:

Let p, q and r be 3 statement variables.

(i) Truth table for $\sim p \vee \sim q$, $\sim(p \vee q)$ and $\sim(p \wedge q)$

p	q	$\sim p$	$\sim q$	$\sim p \vee \sim q$	$p \vee q$	$\sim(p \vee q)$	$p \wedge q$	$\sim(p \wedge q)$
T	T	F	F	F	T	F	T	F
T	F	F	T	T	T	F	F	T
F	T	T	F	T	T	F	F	T
F	F	T	T	T	F	T	F	T

- (a) $\sim p \vee \sim q$ and $\sim(p \vee q)$ are not logically equivalent because when p is true and q is false, $\sim p \vee \sim q$ is true, but $\sim(p \vee q)$ is false.
- (b) $\sim p \vee \sim q$ and $\sim(p \wedge q)$ are logically equivalent because their truth values agree for all possible combinations of the truth values of p and q .

(ii) Truth table for $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$

p	q	r	$q \wedge r$	$p \vee (q \wedge r)$	$p \vee q$	$p \vee r$	$(p \vee q) \wedge (p \vee r)$
T	T	T	T	T	T	T	T
T	T	F	F	T	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	T	T	T	T
F	T	T	T	T	T	T	T
F	T	F	F	F	T	F	F
F	F	T	F	F	F	T	F
F	F	F	F	F	F	F	F

$p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ are logically equivalent because their truth values agree for all possible combinations of the truth values of p, q and r .

Negations of conjunction and disjunction : De Morgan's Law

<u>De Morgan's Law:</u>	$\sim (p \wedge q) \equiv \sim p \vee \sim q$	(See Example 1.2.1(b).)
	$\sim (p \vee q) \equiv \sim p \wedge \sim q$	(Try to show this!)

Example 1.2.2:

- (i) John is smart but lazy.
Negation: John is not smart or not lazy.
- (ii) $\sqrt{2} \leq 2 < \pi$.
Negation: $2 < \sqrt{2}$ or $2 \geq \pi$.

Laws of Logical Equivalences

Sometime, use truth table to show two propositions are logically equivalent is lengthy and tedious. So, we may solve the problem by using some Laws of Logical Equivalences.

Theorem 1.2.1: Laws of Logical Equivalences

Given any statement variables p , q , and r , a tautology t and a contradiction c , the following logical equivalences hold:

1. Commutative laws: $p \wedge q \equiv q \wedge p$, $p \vee q \equiv q \vee p$
2. Associative laws: $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$, $(p \vee q) \vee r \equiv p \vee (q \vee r)$
3. Distributive laws: $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$, $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
4. Identity laws: $p \wedge t \equiv p$, $p \vee c \equiv p$
5. Negation laws: $p \vee \sim p \equiv t$, $p \wedge \sim p \equiv c$
6. Double negation law: $\sim(\sim p) \equiv p$
7. Idempotent laws: $p \wedge p \equiv p$, $p \vee p \equiv p$
8. De Morgan's laws: $\sim(p \wedge q) \equiv \sim p \vee \sim q$, $\sim(p \vee q) \equiv \sim p \wedge \sim q$
9. Universal bound laws: $p \vee t \equiv t$, $p \wedge c \equiv c$
10. Absorption laws: $p \vee (p \wedge q) \equiv p$, $p \wedge (p \vee q) \equiv p$
11. Negation of t and c : $\sim t \equiv c$, $\sim c \equiv t$

Example 1.2.3:

Simply the following:

- (i) $p \vee (p \wedge q)$ (ii) $\sim(p \vee q) \vee (\sim p \wedge q)$.

1.3 Conditional and Biconditional Statements

Definition 1.3.1 : Conditional Statement (Implication) $p \rightarrow q$

Let p and q be statement variables. The statement “*If p then q* ” or “ *p implies q* ” is called a **conditional statement** or **implication**. It is denoted by $p \rightarrow q$.

The statement p is called the **hypothesis** (or **antecedent**) and the statement q is called the **conclusion** (or **consequent**).

It is false, when p is true and q is false; otherwise it is true.

Truth table for $p \rightarrow q$

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Remark:

(1) A variety of terminology is used to express $p \rightarrow q$ as given below:

If p then q , q if p ,
 p implies q , q when p (q whenever p) ,
 p only if q , q follows from p ,
 p is sufficient for q , **q is necessary for p ,**
 q unless $\sim p$.

Example 1.3.1:

Consider the statement:

“If a number is divisible by 4, then it is an even number.”

A number is being divisible by 4 is sufficient for it to be even.

A number is even is necessary for it is being divisible by 4.

When the word “unless” is used, the statement becomes

“A number is even unless it is not divisible by 4.”

- (2) If the hypothesis p is false, $p \rightarrow q$ say nothing about what will happen. In this case, we consider $p \rightarrow q$ is true regardless of its conclusion.
 $p \rightarrow q$ is true because of p is false is often called **true by default**.
- (3) A true conditional statement does not mean a conditional statement with true conclusion.

Example 1.3.2:

- (a) If 2 or $\sqrt{2}$ is negative, then $(\sqrt{2} - 2)$ is negative.
- (b) If 2 or $\sqrt{2}$ is negative, then $(\sqrt{2} + 2)$ is negative.

Both of the above conditional statements are true by default. However, statement (a) has a true conclusion but statement (b) has a false conclusion.

- (4) In ordinary English, there is a cause-and-effect relation between hypothesis p and conclusion q . However, in logic we use conditional statements in a more general sense that is customary.

Examples 1.3.3: Examples of conditional statement:

- (a) If Mary studies hard, then she will complete her course this year.
- (b) If $3 < 5$, then Kuala Lumpur is the Capital of Malaysia.

Example 1.3.4:

Determine the hypothesis and conclusion of the following conditional statements. Then determine the truth-value of each conditional statement.

- (1) The moon is square only if the sun rises in the East.
- (2) 1 and 3 are prime if 1 multiply 3 is prime.
- (3) $(\sin \pi)(\cos \pi) = 0$ when $\sin \pi = 0$ or $\cos \pi = 0$.
- (4) February has 30 days implies March has 31 days.

The Negation, Contrapositive, Converse and Inverse of $p \rightarrow q$

Definition 1.3.2:

Suppose a conditional statement $p \rightarrow q$ is given.

- (1) The **negation** of $p \rightarrow q$ is $p \wedge \sim q$.
- (2) The **contrapositive** of $p \rightarrow q$ is $\sim q \rightarrow \sim p$.
- (3) The **converse** of $p \rightarrow q$ is $q \rightarrow p$.
- (4) The **inverse** of $p \rightarrow q$ is $\sim p \rightarrow \sim q$.

Example 1.3.5:

Truth table for $\sim(p \rightarrow q)$ and $p \wedge \sim q$

p	q	$\sim q$	$(p \rightarrow q)$	$\sim(p \rightarrow q)$	$p \wedge \sim q$
T	T	F	T	F	F
T	F	T	F	T	T
F	T	F	T	F	F
F	F	T	T	F	F

$$\therefore \sim(p \rightarrow q) \equiv p \wedge \sim q.$$

The negation of “If p then q ” is logically equivalent to “ p and not q ”.

Do not write the negation of an “if-then” statement as another “if-then” statement.

Example 1.3.6:

- (a) If 2 and 3 are prime numbers then $2 + 3$ is prime.

Negation: _____

Contrapositive: _____

Converse: _____

Inverse: _____

- (b) If $x^2 = 2$, then $x = \pm\sqrt{2}$.

Negation: _____

Contrapositive: _____

Converse: _____

Inverse: _____

Definition 1.3.3: Biconditional Statement

Let p and q be statement variables.

The proposition $(p \rightarrow q) \wedge (q \rightarrow p)$
 $(p \text{ only if } q) \text{ and } (p \text{ if } q)$
 $(p \text{ is sufficient for } q) \text{ and } (p \text{ is necessary for } q)$

is called the **biconditional of p and q** .

This is read “ p if and only if q ” or “ p is a necessary and sufficient condition for q ”) and is denoted $p \leftrightarrow q$.

It is true if both p and q have the same truth values and it is false if p and q have the opposite truth values.

Truth table for $p \leftrightarrow q$

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Example 1.3.7: Determine the truth values of the following statements.

(1) 2 is prime if and only if it is multiple of 2.

(2) $\sqrt{2}$ is negative if and only if 2 is negative.

Remark:

We can use a biconditional statement to determine whether 2 propositions P and Q are logically equivalent or not.

$P \equiv Q$ if, and only if $P \leftrightarrow Q$ is a tautology.

Logical equivalences of conditional statements

Theorem 1.3.1: Let p and q be statement variables.

(1) $p \rightarrow q \equiv \sim p \vee q$

(2) $p \rightarrow q \equiv \sim q \rightarrow \sim p$

(A conditional statement and its contrapositive are logically equivalent)

(3) $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

Proof:(1) Truth table for $p \rightarrow q \leftrightarrow \sim p \vee q$

p	q	$p \rightarrow q$	$\leftrightarrow$	$\sim p \vee q$
T	T	T	T	T
T	F	F	T	F
F	T	T	T	T
F	F	T	T	T

Since $p \rightarrow q \leftrightarrow \sim p \vee q$ is a tautology, so $p \rightarrow q \equiv \sim p \vee q$.

(2) Truth table for $p \rightarrow q \leftrightarrow \sim q \rightarrow \sim p$

p	q	$p \rightarrow q$	$\leftrightarrow$	$\sim q \rightarrow \sim p$
T	T	T	T	T
T	F	F	T	F
F	T	T	T	T
F	F	T	T	T

Since $p \rightarrow q \leftrightarrow \sim q \rightarrow \sim p$ is a tautology, so $p \rightarrow q \equiv \sim q \rightarrow \sim p$.

Only if

To say “ p only if q ” means that p can take place **only** if q takes place **also**. That is, if q does not take place, then p cannot take place.

Definition 1.3.4: Only if

Let p and q be statement variables. “ p only if q ” means “*If not q then not p* ”.

Example 1.3.8:

The function f is differentiable only if f is continuous.

This conditional statement means if the function f is not continuous then it is not differentiable.

Example 1.3.9:

Show that (i) $(p \wedge q) \rightarrow (p \vee q)$ is a tautology;

(ii) $p \rightarrow (q \wedge r) \equiv (p \rightarrow q) \wedge (p \rightarrow r)$.

1.4 Predicates and Quantifiers

Let $P(x, y)$ denote " $x + y > 3$ ".

It is not necessarily true or false unless x and y are replaced by particular numbers.

So, the sentence $P(x, y): x + y > 3$ is not a statement.

However, when the variable x and y are replaced by particular elements a and b , $P(a, b)$ is a statement.

Example: $P(5, 1): 5 + 1 > 3$ is a true statement;
 $P(-2, 2): -2 + 2 > 3$ is a false statement.

In this case, $P(x, y)$ is called a *predicate*.

Definition 1.4.1: Predicate

A sentence is a *predicate* (or *propositional function*) if

- (1) it contains a finite number of *predicate variables* (eg. $x_1, x_2, \dots, x_n$), and
- (2) it is not a statement, but it becomes a statement when particular values are substituted for the variables.

The *universe of discourse* or *domain* D of a predicate variable is the set of all values that may be substituted in place of the variable.

Example 1.4.1:

- (a) Let $P(x): x$ is an animal and domain = {apple, cat, computer, elephant, horse}.

$P(\text{cat})$, $P(\text{elephant})$ and $P(\text{horse})$ are true statements.

$P(\text{apple})$ and $P(\text{computer})$ are false statements.

- (b) Let $P(x, y): x \geq y$ and domain = $\{(x, y) \mid x = 1, 2, \text{ or } 3, \text{ and } y = 3 \text{ or } 4\}$.

$P(3, 3)$ is the only true statement.

$P(1, 3)$, $P(1, 4)$, $P(2, 3)$, $P(2, 4)$ and $P(3, 4)$ are false statements.

- (c) Let $P(x): x$ is prime $\rightarrow x$ is odd. What is the truth value of each of the following statements?

(i) $P(17)$

(ii) $P(2)$

(iii) $P(3)$

(iv) $P(0)$

- (d) Let $P(x, y): x^2 - y^2 = 0 \leftrightarrow x = y$. What is the truth value of each of the following statements?

(i) $P(1, 1)$

(ii) $P(1, -1)$

(iii) $P(1, 0)$

Quantifiers: $\forall$ and $\exists$

Let $P(x)$ be a predicate. Beside substitute a particular value to x , another way to obtain statements from predicate is to add *quantifiers*.

Quantifiers are words that refer to quantities such as “all” or “some” and tell for how many elements a given predicate is true.

(1) Universal quantifier $\forall$

Definition 1.4.2: Universal statement

Let $P(x)$ be a predicate. A ***universal statement*** is a statement of the form

$$\forall x P(x)$$

means “ $P(x)$ is true for **all** values of x in the domain”.

Remark:

- (1) $\forall$ is called ***universal quantifier***.
- (2) $\forall x P(x)$ is read as “**for all** $x P(x)$ ”, “**for every** $x P(x)$ ” or “**for any** $x P(x)$ ”.
- (3) A universal statement is true if, and only if $P(x)$ is true for **all** x in the domain.
- (4) A universal statement is false if, and only if $P(x)$ is false for **at least one** x in the domain. A value for x for which $P(x)$ is false is called a **counterexample** to the universal statement.

Example 1.4.2:

Determine the truth value of the following universal statement. If the universal statement is false, suggest a counterexample to the universal statement.

- (a) $\forall x \in \{-1, 0, 3, 4, 5\}, 2x$ is an even integer.
- (b) $\forall x \in \{x \mid 0 \leq x \leq 1\}, x < 5x$.
- (c) $\forall x \in \mathbb{Z}, (x + 1)$ is nonnegative.

(2) Existential quantifier $\exists$

Definition 1.4.3: Existential statement

Let $P(x)$ be a predicate. An *existential statement* is a statement of the form

$$\exists x P(x)$$

means “**There exists an** element x in the domain such that $P(x)$ is true”.

Remark:

- (1) $\exists$ is called *existential quantifier*.
- (2) $\exists x P(x)$ is also read as “**There is an** x such that $P(x)$ ” or “**There is at least one** x such that $P(x)$ ” or “**For some** $x P(x)$ ”.
- (3) An existential statement is true if, and only if $P(x)$ is true for **at least one** x in the domain.
- (4) An existential statement is false if, and only if $P(x)$ is false for **all** x in the domain.
- (5) Unique Existence $\exists!$
 $\exists! x P(x)$ is read as “**There exists a unique** x such that $P(x)$ ”.

Example 1.4.3:

Determine the truth value of the following existential statement.

- (1) $\exists x \in R$ such that $x \leq x + 1$.
- (2) $\exists x \in Z$ such that $x^2 = 2$.
- (3) $\exists! x \in R$ such that $2x + 3 = 5x - 7$.

Rules for Negating Statements with One Quantifier

(1) Negation of universal statement

$$\sim (\forall x P(x)) \equiv \exists x \sim P(x).$$

The negation of a universal statement “all are” is logically equivalent to an existential statement “some are not” or “at least one is not”.

(2) Negation of existential statement

$$\sim (\exists x P(x)) \equiv \forall x \sim P(x)$$

The negation of an existential statement “some are” or “at least one is” is logically equivalent to a universal statement “all are not” or “No___”.

Example 1.4.4:

Translate each of the following statements into symbolic form using quantifiers. Then write their negations in logical expression form and in English. In each case, the assumed domain is given in parentheses.

- (1) Every prime number is odd. (Domain: All real numbers)

Symbolic form: _____

Negation: _____

- (2) No triangle has four sides. (Domain: All polygons)

Symbolic form: _____

Negation: _____

- (3) Some Malaysian eats cheeseburgers but does not eat cheesecake. (Domain: All Malaysians)

Symbolic expression form: _____

Negation: _____

Statements containing multiple quantifiers

A multiply quantified statement is a statement contains more than one quantifiers.

Example 1.4.5:

Translate each of the following statements into symbolic form using quantifiers. Which of the following statements is true?

- (1) Given any positive number, there is another positive number that is smaller than the given number.

Symbolic form: $\forall x \in R^+, \exists y \in R^+ \text{ such that } y < x.$

- (2) There is a real number x such that for every real number y , the sum of the x and y is zero.

Symbolic form: $\exists x \in R \text{ such that } \forall y \in R, x + y = 0.$

1.5 Arguments

Definition 1.5.1: Argument

An **argument** is An assertion that given a set of propositions $P_1, P_2, \dots, P_n$ called **premises**, yields (as a consequence) another proposition Q called the **conclusion**. Such an argument is denoted by

$$\begin{array}{c} P_1 \\ P_2 \\ \vdots \\ P_n \end{array} \quad \text{or} \quad P_1, P_2, \dots, P_n \quad Q,$$

$$\therefore Q \quad (\therefore \text{ is read as therefore})$$

which means $(P_1 \wedge P_2 \wedge \dots \wedge P_n) \rightarrow Q$.

Example 1.5.1: Examples of argument

- (a) If it rains, I drive to work. (b) If Ali studies at UM, he stays in PJ.
I didn't drive to work. Ali doesn't stay in PJ or he drives to campus.
 $\therefore$ It did not rain. Ali doesn't drive to campus.
 $\therefore$ Ali doesn't study at UM.

Definition 1.5.2: Valid and fallacy argument

An argument is **valid** provided that if all premises $P_1, P_2, \dots, P_n$ are true, then the conclusion Q must be also true; otherwise the argument is **invalid** (or **fallacy**).

Theorem 1.5.1: The argument $P_1, P_2, \dots, P_n \quad Q$ is valid
if and only if $(P_1 \wedge P_2 \wedge \dots \wedge P_n) \rightarrow Q$ is a tautology.

Example 1.5.2:

Test the validity of the argument in Example 1.5.1(a).

Solution: Let p : It rains; q : I drive to work.

Symbolically, the argument is denoted by $[(p \rightarrow q) \wedge \sim q] \rightarrow \sim p$.

Truth table for $[(p \rightarrow q) \wedge \sim q] \rightarrow \sim p$						
p	q	$[(p \rightarrow q) \wedge \sim q] \rightarrow \sim p$				
T	T	T	F	F	T	F
T	F	F	F	T	T	F
F	T	T	F	F	T	T
F	F	T	T	T	T	T

The argument is valid since it is a tautology.

Table 1.5.1: Rules of Inference (Some valid argument forms)

(1) <u>Law of Detachment (Modus Ponens)</u> $\begin{array}{c} p \\ \underline{p \rightarrow q} \\ \therefore q \end{array}$	(2) <u>Modus Tollens</u> $\begin{array}{c} p \rightarrow q \\ \sim q \\ \therefore \sim p \end{array}$
(3) <u>Disjunctive Addition</u> $\begin{array}{ccc} p & \text{or} & q \\ \therefore p \vee q & & \therefore p \vee q \end{array}$	(4) <u>Conjunctive Simplification</u> $\begin{array}{ccc} \underline{p \wedge q} & \text{or} & \underline{p \wedge q} \\ \therefore p & & \therefore q \end{array}$
(5) <u>Disjunctive Syllogism</u> $\begin{array}{ccc} p \vee q & \text{or} & p \vee q \\ \sim p & & \sim q \\ \therefore q & & \therefore p \end{array}$	(6) <u>Conjunction</u> $\begin{array}{c} p \\ q \\ \therefore p \wedge q \end{array}$
(7) <u>Law of the Syllogism</u> $\begin{array}{c} p \rightarrow q \\ \underline{q \rightarrow r} \\ \therefore p \rightarrow r \end{array}$	(8) <u>Resolution</u> $\begin{array}{c} p \vee q \\ \underline{\sim p \vee r} \\ \therefore q \vee r \end{array}$

Example 1.5.3:

Use rules of inferences to prove that the following argument is valid.

I will enroll in SJES1210 or SJES1221 in year 1.
I didn't enroll in SJES1221 in year 1.
 $\therefore$ I enrolled in SJES1210 in year 1.

Solution:

Let p : I enrolled in SJES1210,
 q : I enrolled in SJES1221.

The argument is denoted by
$$\begin{array}{c} p \vee q \\ \underline{\sim q} \\ \therefore p \end{array}$$

<u>Steps</u>	<u>Reasons</u>
(1) $p \vee q$	Premise
(2) $\sim p \rightarrow q$	Logical equivalence form of (1)
(3) $\sim q \rightarrow p$	Contrapositive of (2)
(4) $\sim q$	Premise
$\therefore p$	(3), (4) and Modus Ponens

Remark:

When an argument is valid and its premises are true, the truth of the conclusion is said to be inferred or deduced from the truth of the premises.

Notice that we are not saying that the conclusion is true; we only saying the conclusion is true provided the premises (hypotheses) all are true.

Example 1.5.4:

Use rules of inferences to prove that the following argument is valid.

If the band could not perform or the refreshments were not delivered on time, then the party would have been cancelled and Amy would have been angry. Refunds would have had to be made unless the party did not cancel. No refunds were made.
Therefore, the band could perform.

Solution:

Let p : The band could perform,

q : The refreshments were delivered on time,

r : The party was cancelled,

u : Amy was angry,

w : Refunds had to be made.

The argument is denoted by

$$\begin{array}{l} (\sim p \vee \sim q) \rightarrow (r \wedge u) \\ w \vee \sim r \\ \hline \sim w \\ \hline \therefore p \end{array}$$

<u>Steps</u>	<u>Reasons</u>
(1) $w \vee \sim r$	Premise
(2) $\sim w$	Premise
(3) $\sim r$	(1), (2) and Disjunctive Syllogism
(4) $\sim r \vee \sim u$	(3) and Disjunctive Addition
(5) $\sim (r \wedge u)$	(4) and DeMorgan's Law
(6) $(\sim p \vee \sim q) \rightarrow (r \wedge u)$	Premise
(7) $\sim (\sim p \vee \sim q)$	(5), (6) and Modus Tollens
(8) $p \wedge q$	(7), DeMorgan's Law and Law of Double Negation
$\therefore p$	(8) and Conjunctive Simplification

1.6 Mathematical Induction

Consider the following problem:

Can you write down a general rule for the sequence: $1, 1+8, 1+8+27, 1+8+27+64, \dots$?

We can look at the above sequence as: $a_1 = 1^3,$

$$a_2 = 1^3 + 2^3 = a_1 + 2^3,$$

$$a_3 = 1^3 + 2^3 + 3^3 = a_2 + 3^3,$$

$$a_4 = 1^3 + 2^3 + 3^3 + 4^3 = a_3 + 4^3,$$

$\vdots$

Clearly, each number in the sequence can be obtained from the preceding one by adding the appropriate perfect cube.

Thus, the sequence can be defined by $a_1 = 1, a_n = a_{n-1} + n^3$ for $n > 1$.

We say $a_n, n > 1$, has been defined **inductively**, or **recursively**.

An **inductive definition** gives a specific definition in the first case (a_1 in this example), and then gives a rule for computing any case from its predecessor (or, in some cases, from more than one predecessor.)

The Principle of Mathematical Induction

Mathematical induction is a method for proving statement about the positive integers. Let $P(n)$ be such a predicate. The purpose of an induction proof is to show that $P(n)$ is true for **every** positive integer n .

A proof by mathematical induction is in some ways similar to defining sequences or other mathematical objections inductively.

In an inductive definition, we specific the value for the first case, and tell how to get each subsequent case from its predecessor. In a proof mathematical induction, we prove the first case directly, then show that how to prove each subsequent case under the assumption that its predecessors are true.

Theorem 1.6.1: First Principle of Mathematical Induction

Let $P(n)$ be a predicate that is defined for positive integers n . Suppose that

- (1) $P(1)$ is true.
- (2) For all integers $k \geq 1$, if $P(k)$ is true, then $P(k + 1)$ is true.

Then $P(n)$ is true for all positive integers n .

Remark:

Proving a statement by induction is a two-steps process. The Step (1) is the **initial step** and the Step (2) is the **inductive step**. The assumption $P(k)$ is true in the Step (2) is called the **inductive hypothesis**.

Example 1.6.1: Prove that, for any $n \in \mathbb{Z}^+$, $\sum_{i=1}^n i = 1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$.

Solution:

Let $P(n): \sum_{i=1}^n i = 1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$.

1. When $n=1$, $\sum_{i=1}^1 i = 1 = 1(1+1)/2$. So, $P(1)$ is true.

2. Assume that $P(k)$ is true **for some** k with $k \geq 1$.

That is, $P(k): \sum_{i=1}^k i = 1 + 2 + \cdots + k = \frac{k(k+1)}{2}$ is true **for some** k .

When $n = k+1$, $\sum_{i=1}^{k+1} i = 1 + 2 + \cdots + k + (k+1) = (k+1) + \sum_{i=1}^k i$

$$= \frac{k(k+1)}{2} + (k+1) = \frac{k(k+1) + 2(k+1)}{2} = \frac{(k+1)(k+2)}{2}.$$

So, $P(k+1): \sum_{i=1}^{k+1} i = \frac{k(k+2)}{2}$ is true.

Therefore, $P(n)$ is true for all positive integers n .

Example 1.6.2: Prove that, for any $n \in \mathbb{Z}^+$, $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$.

Solution:

Let $P(n): \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$.

1. When $n=1$, $\sum_{i=1}^1 i^2 = 1^2 = \frac{1(1+1)(2+1)}{6}$. So, $P(1)$ is true.

2. Assume that $P(k)$ is true **for some** k with $k \geq 1$.

That is, $P(k): \sum_{i=1}^k i^2 = \frac{k(k+1)(2k+1)}{6}$ is true **for some** k .

When $n = k+1$, $\sum_{i=1}^{k+1} i^2 = 1^2 + 2^2 + \cdots + k^2 + (k+1)^2 = \frac{k(k+1)(2k+1)}{6} + (k+1)^2$

$$= \frac{(k+1)}{6} [k(2k+1) + 6(k+1)] = \frac{(k+1)}{6} (2k^2 + 7k + 6)$$

$$= \frac{(k+1)}{6} (k+2)(2k+3).$$

So, $P(k+1): \sum_{i=1}^{k+1} i^2 = \frac{(k+1)(k+2)(2k+3)}{6}$ is true.

Therefore, $P(n)$ is true for all positive integers n .

Theorem 1.6.2: Extended Principle of Mathematical Induction

Let $P(n)$ be a predicate that is defined for integers $n \geq a$ where a is a fixed integer. Suppose that

- (1) $P(a)$ is true.
 (2) For all integers $k \geq a$, if $P(k)$ is true, then $P(k + 1)$ is true.

Then $P(n)$ is true for all integers $n \geq a$.

Example 1.6.3: Prove that $(n + 1)! > 2^{n+3}$ for all integers $n \geq 5$.

Solution:

Let $P(n)$: $(n + 1)! > 2^{n+3}$.

1. When $n = 5$, $(5 + 1)! = 6! = 720$ and $2^{5+3} = 2^8 = 256$,
 so $P(5)$: $(5 + 1)! > 2^{5+3}$ is true.

2. Assume that $P(k)$ is true *for some* integer k with $k \geq 5$.

That is, $P(k)$: $(k + 1)! > 2^{k+3}$ is true *for some* integer k with $k \geq 5$.

$$\begin{aligned} \text{When } n = k + 1, \quad (k + 1 + 1)! &= (k + 2)! = (k + 2)(k + 1)! \\ &> (k + 2)2^{k+3} = k2^{k+3} + 2^{k+4} > 2^{k+4}. \end{aligned}$$

So, $P(k + 1)$: $(k + 2)! > 2^{k+4}$ is true.

Therefore, $P(n)$ is true for all integer $n \geq 5$.

1.7 The Binomial Theorem

A *binomial expression* is simply the sum of two terms, such as $x + y$. The Binomial Theorem gives the coefficients of the expansion of powers of binomial expression $(x + y)^n$.

Theorem 1.7.1: The Binomial Theorem

Let x and y be variables, and let n be a positive integer. Then

$$(x + y)^n = \sum_{r=0}^n \binom{n}{r} x^r y^{n-r} = \binom{n}{0} y^n + \binom{n}{1} xy^{n-1} + \cdots + \binom{n}{r} x^r y^{n-r} + \cdots + \binom{n}{n} x^n.$$

Remark: $\binom{n}{r} = \frac{n!}{r!(n-r)!}$ (also denoted as nC_r) is called a *binomial coefficient*.

Theorem 1.7.2: Pascal's Identity

Let n and r be positive integers with $n \geq r$. Then $\binom{n+1}{r} = \binom{n}{r-1} + \binom{n}{r}$.

Pascal's Identity is the basis for a geometric arrangement of the binomial coefficients in a triangle called *Pascal's triangle*.

Table 1.7.1: PASCAL'S TRIANGLE

$$\begin{array}{cccccccccccc}
& & & & & & & & & & r=0 \\
& & & & & & & & & & 1 & r=1 \\
n=1 & & & & & & 1 & 1 & & & r=2 \\
n=2 & & & & & 1 & 2 & 1 & & & r=3 \\
n=3 & & & & 1 & 3 & 3 & 1 & & & r=4 \\
n=4 & & & 1 & 4 & 6 & 4 & 1 & & & r=5 \\
n=5 & & 1 & 5 & 10 & 10 & 5 & 1 & & & r=6 \\
n=6 & & 1 & 6 & 15 & 20 & 15 & 6 & 1 & & r=7 \\
n=7 & & 1 & 7 & 21 & 35 & 35 & 21 & 7 & 1 & r=8 \\
n=8 & 1 & 8 & 28 & 56 & 70 & 56 & 28 & 8 & 1 & r=9 \\
n=9 & 1 & 9 & 36 & 84 & 126 & 126 & 84 & 36 & 9 & 1 \\
& \vdots & & & & & & & & &
\end{array}$$

Example 1.7.1:

Expand the following (i) $(x - 3)^5$ (ii) $(x - 2y)^4$.

Solution:

$$\begin{aligned} \text{(i)} \quad (x-3)^5 &= {}^5C_0(-3)^5 + {}^5C_1x(-3)^4 + {}^5C_2x^2(-3)^3 + {}^5C_3x^3(-3)^2 + {}^5C_4x^4(-3) + {}^5C_5x^5 \\ &= -243 + 405x - 270x^2 + 90x^3 - 15x^4 + x^5. \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad (x-2y)^4 &= {}^4C_0(-2y)^4 + {}^4C_1x(-2y)^3 + {}^4C_2x^2(-2y)^2 + {}^4C_3x^3(-2y) + {}^4C_4x^4 \\ &= 16y^4 - 32xy^3 + 24x^2y^2 - 8x^3y + x^4. \end{aligned}$$

Example 1.7.2:

(i) Find the coefficient of x^5y^2 in the expansion of $(x+y)^7$.

(ii) Find the coefficient of $a^5 b^2$ in the expansion of $(2a - 3b)^7$.

Solution:

(i) The coefficient of $x^5 y^2$ in the expansion of $(x + y)^7$ is ${}^7C_5 = 21$.

(ii) The coefficient of $a^5 b^2$ in the expansion of $(2a - 3b)^7$ is ${}^7C_5(2)^5(-3)^2 = 6048$.

Chapter 2 Set Theory and Functions

2.1 Sets

A **set** is any well-defined list or collection of objects. The objects in a set are called **elements**, or **members**.

Remark:

- (1) Putting the elements of a set in brackets $\{ \}$ is a standard notation.
For example: $A = \{1, 2, 3\}$ is a set with three elements.
- (2) $a \in A$ is read as “ a is in A ”, means the object a is an element of the set A .
- (3) $a \notin A$ is read as “ a is not in A ”, means the object a is not an element of A .
- (4) A set with no element is called **the empty** or **null set**. It is denoted by $\emptyset$ or $\{ \}$.
- (5) $\{x|P(x)\}$ is the set that satisfies the property $P(x)$. The symbol $|$ is read as “such that”.
 $\{x \in X | P(x)\}$ is read as “the set of all x in X such that $P(x)$ ”.

Example 2.1.1:

Which of the following sets are empty?

- (i) $A = \{0\}$ (ii) $B = \{x \in \mathbb{R} | x^2 < 0\}$ (iii) $D = \{\emptyset\}$

Definition 2.1.1:

Two sets A and B are **equal**, written $A = B$, if and only if they have the same elements.

Example 2.1.2:

Which of the following sets are equal?

$$A = \{1, 2, 3, 4\}, \quad B = \{x \in \mathbb{R} | x^2 - 5x + 6 = 0\},$$

$$C = \{x \in \mathbb{R} | x \text{ is positive}\}, \quad D = \{x \in \mathbb{Z} | x > 0\},$$

$$E = \{x \in A | x \text{ is prime}\}, \quad F = \{x \in \mathbb{R} | 1 \leq x < 5\}.$$

Subsets

Definition 2.1.2: Subsets

Let A and B be any two sets.

We say A is a **subset** of B , and write $A \subseteq B$, if and only if every element of A is also an element of B .

If A is a subset of B and $A \neq B$, we say A is a **proper subset** of B , and write $A \subset B$.

A is **not a subset** of B if at least one element of A does not belong to B , and write $A \not\subseteq B$.

Remark:

1. Symbolically, $A \subseteq B \Leftrightarrow \forall x$, if $x \in A$ then $x \in B$.
2. The empty set $\emptyset$ is a subset of any set A , that is, $\emptyset \subseteq A$.
3. If A is any set, then $A \subseteq A$.
4. $A = B$ where A and B are two sets, if and only if, $A \subseteq B$ and $B \subseteq A$.
5. If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.

Example 2.1.3: Distinction Between $\in$ and $\subseteq$

Which of the following statements are true?

- | | |
|---|--|
| (i) $16 \in \{x \mid x \text{ is even}\}$ | (ii) $\{0\} \subseteq \emptyset$ |
| (iii) $\{4, 5\} \subseteq \{x \in \mathbb{N} \mid x \leq 5\}$ | (iv) $\{1, 2\} \in \{1, 2, 3\}$ |
| (v) $\{1\} \in \{\{1\}, \{2\}\}$ | (vi) $\{1, 2\} \subseteq \{\{1\}, \{2\}, \{1, 2\}\}$ |

Definition 2.1.3: Power set

The set of all subsets of a set A is called the **power set** of A , and is denoted by $\mathcal{P}(A)$.

Example 2.1.4:

If $A = \{0, 1\}$, the power set of $A = \mathcal{P}(A) = \{\emptyset, \{0\}, \{1\}, \{0, 1\}\}$.

If $B = \{\emptyset, \{1\}\}$, the power set of $B = \mathcal{P}(B) = \{\emptyset, \{\emptyset\}, \{\{1\}\}, \{\emptyset, \{1\}\}\}$.

If $C = \{1, 2, 3\}$, the power set of $C = \mathcal{P}(C) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, C\}$.

Operations on sets

In any discussion involving sets, we work within a **universal set** U , which contains all the elements of all the sets in the discussion.

Definition 2.1.4: Unions, Intersections and Complements

Let A and B be sets.

The **union** of A and B , denoted $A \cup B$, is $\{x \mid x \in A \text{ or } x \in B\}$.

The **intersection** of A and B , denoted $A \cap B$, is $\{x \mid x \in A \text{ and } x \in B\}$.

If $A \cap B = \emptyset$, then A and B are said to be **disjoint** or **non-intersecting**.

The **complement** of A , denoted A' (or $\bar{A}$), is $\{x \mid x \notin A\}$.

The **difference** of A and B (or **relative complement** of B in A), denoted $A - B$, is $\{x \mid x \in A \text{ but } x \notin B\}$.

Example 2.1.5:

Let $\mathbb{N}$ be the universal set, $A = \{x \in \mathbb{N} \mid x \leq 6\}$ and $B = \{x \in \mathbb{N} \mid x \text{ is a positive factor of } 12\}$.

- Find (i) $A \cup B$ (ii) $A \cap B$ (iii) A'
 (iv) $(A \cup B)'$ (v) $A - B$ (vi) $(A - B) \cap (A \cup B)'$

Solution:

$A = \{1, 2, 3, 4, 5, 6\}$ and $B = \{1, 2, 3, 4, 6, 12\}$.

- (i) $A \cup B = \{1, 2, 3, 4, 5, 6, 12\}$ (ii) $A \cap B = \{1, 2, 3, 4, 6\}$
 (iii) $A' = \{x \in \mathbb{N} \mid x \geq 7\}$ (iv) $(A \cup B)' = \{x \in \mathbb{N} \mid x \geq 7 \text{ and } x \neq 12\}$
 (v) $A - B = \{5\}$ (vi) $(A - B) \cap (A \cup B)' = \emptyset$

Theorem 2.1.1: The Algebra of Sets, Basic Properties

Let A , B , and C be sets of a universal set U . Then all the following equalities:

1. Idempotent laws: $A \cap A = A$, $A \cup A = A$
2. Commutative laws: $A \cap B = B \cap A$, $A \cup B = B \cup A$
3. Associative laws: $(A \cap B) \cap C = A \cap (B \cap C)$, $(A \cup B) \cup C = A \cup (B \cup C)$
4. Distributive laws: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$, $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
5. Double complement law: $(A')' = A$
6. De Morgan's laws: $(A \cap B)' = A' \cup B'$, $(A \cup B)' = A' \cap B'$
7. Absorption laws: $A \cup (A \cap B) = A$, $A \cap (A \cup B) = A$
8. Properties of $\emptyset$: $A \cap \emptyset = \emptyset$, $A \cup \emptyset = A$
9. Properties of U : $A \cap U = A$, $A \cup U = U$

Finite Sets and Counting Principle**Definition 2.1.5: Finite sets**

A set is said to be **finite** if it contains exactly m distinct elements where m is a nonnegative integer. Otherwise, a set is said to be **infinite**.

If A is a finite set, then the number of elements of A is denoted by $n(A)$ or $|A|$.

Example 2.1.6:

Let $A = \{x \in \mathbb{N} \mid x \leq 5\}$, $B = \{x \mid x \text{ is even}\}$ and $C = \{x \in \mathbb{R} \mid 1 \leq x \leq 5\}$.

- (i) $A = \{1, 2, 3, 4, 5\}$ is a finite set and $|A| = 5$.
 (ii) B and C are infinite sets.

Theorem 2.1.2:

If A and B are finite sets, then $(A \cup B)$ and $(A \cap B)$ are also finite, and

$$n(A \cup B) = n(A) + n(B) - n(A \cap B).$$

We can apply Theorem 2.1.2 to obtain a formula for three finite sets.

Theorem 2.1.3:

If A , B and C are finite sets, then $(A \cup B \cup C)$ and $(A \cap B \cap C)$ are also finite, and

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C).$$

Example 2.1.7:

Suppose that 100 of 120 form five students in a secondary school take at least one of the subjects: mathematics, physics and chemistry. Also suppose:

65 take mathematics, 45 take physics, 42 take chemistry and 20 take mathematics and physics, 25 take mathematics and chemistry, and 15 take physics and chemistry.

Determine the number of student who take

- (i) all the three subjects,
- (ii) only two subjects,
- (iii) only one subjects,
- (iv) none of the subjects.

Solution:

Let M = the set of all the student take mathematic,

P = the set of all the student take physics,

C = the set of all the student take chemistry.

Given that $n(M \cup P \cup C) = 100$, $n(M) = 65$, $n(P) = 45$, $n(C) = 42$,

$n(M \cap P) = 20$, $n(M \cap C) = 25$ and $n(P \cap C) = 15$.

- (i) The number of student who take all the three subjects
 $= n(M \cap P \cap C)$
 $= n(M \cup P \cup C) - n(M) - n(P) - n(C) + n(M \cap P) + n(M \cap C) + n(P \cap C)$
 $= 100 - 65 - 45 - 42 + 20 + 25 + 15 = 8.$
- (ii) From the Venn diagram,
the number of student who take only two subjects
 $= 12 + 17 + 7 = 36.$
- (iii) From the Venn diagram,
the number of student who take only one subjects
 $= 28 + 18 + 10 = 56.$
- (iv) The number of student who take none of the subjects $= 120 - 100 = 20.$

Sets of numbers

In mathematics, we deal with many different sets of objects; some important sets are the set of all real numbers and its subsets.

Standard notation is used for the following sets of numbers:

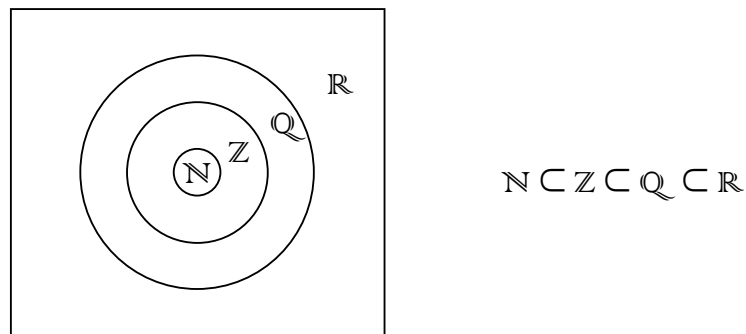
$\mathbb{N}$: the set of all **nature** numbers, that is $\{1, 2, 3, \dots\}$.

$\mathbb{Z}$: the set of all **integers**, that is $\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$.

$\mathbb{Q}$: the set of all **rational** numbers, that is $\left\{ \frac{p}{q} \mid p, q \in \mathbb{Z}, q \neq 0 \right\}$.

$\mathbb{Q}'$: the set of all **irrational** numbers, that is $\{x \in \mathbb{R} \mid x \notin \mathbb{Q}\}$.

$\mathbb{R}$: the set of all **real** numbers.



Remark:

1. The set of all **positive** real numbers = $\{x \in \mathbb{R} \mid x > 0\}$, and the set of all **negative** real numbers = $\{x \in \mathbb{R} \mid x < 0\}$.
Zero is neither positive nor negative.
2. Real numbers can be added, subtracted, multiplied and divided (except by 0).

Example 2.1.8:

Which of the following statements are true?

- | | |
|--|--|
| (i) $-7 \in \mathbb{N}$ | (ii) $\pi \in \mathbb{Q}$ |
| (iii) $\{0, 0.25\} \subset \mathbb{Q}$ | (iv) $\sqrt{-3} \in \mathbb{Q}'$ |
| (iv) $\{-\sqrt{4}\} \in \mathbb{Z}$ | (iii) $\{0, \pi, \pi^2\} \subset \mathbb{R}$ |

2.2 Relations

Definition 2.2.1 : Product set or Cartesian product

For two nonempty sets A and B , the **product set** or **Cartesian product** of A and B is the set of all ordered pairs (a, b) with $a \in A$ and $b \in B$.

Notation: $A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$

Remark:

1. For any two finite, nonempty sets A and B , $n(A \times B) = n(A) \cdot n(B)$
2. In general, $A \times B \neq B \times A$. (The pairs $(a, b) \neq (b, a)$.)
3. $A \times A$ often written as A^2 .

Example 2.2.1:

Let $A = \{a, b\}$ and $B = \{1, 2, 3, 4\}$. Find (i) $A \times B$ (ii) $B \times A$ (iii) A^2 .

Solution:

- (i) $A \times B = \{(a, 1), (a, 2), (a, 3), (a, 4), (b, 1), (b, 2), (b, 3), (b, 4)\}$.
- (ii) $B \times A = \{(1, a), (1, b), (2, a), (2, b), (3, a), (3, b), (4, a), (4, b)\}$.
- (iii) $A^2 = A \times A = \{(a, a), (a, b), (b, a), (b, b)\}$.

Relations

The notation of a relation between two sets of objects is quite common and important. For human being, throughout life, we experience a wide variety of personal relationships. The words such as “father of”, “son of”, “friend of”, etc are about relationships between two persons. For two objects, typical comparisons using such words as “bigger than”, “better than”, “faster than”, etc. In this section, we study that elements within a set or different sets often have a special connection with one another that can be described as a relation. Here, we shall focus our attention to **binary relation**, a relation between two elements.

Definition 2.2.2 : Binary relation

Let A and B be two nonempty sets and let $a \in A$, $b \in B$.

A **binary relation R from A to B** is a subset of $A \times B$. (i.e. $R \subseteq A \times B$.)

If $(a, b) \in R$, we say that **a is R -related to b** (or, **a is related to b by the relation R**) and denote this by **aRb** . ($aRb \leftrightarrow (a, b) \in R$.)

If $(a, b) \notin R$, then **a is not R -related to b** and denote this by **$a \not R b$** .

A **binary relation R on A** is a subset of $A \times A$. (i.e. $R \subseteq A \times A$.)

Example 2.2.2:

Let $A = \{\text{Kuala Lumpur (k), Jakarta (j), Petaling Jaya (p), Shanghai (s), London (l)}\}$ and $B = \{\text{Malaysia (m), Thailand (t), China (c), Indonesia (i)}\}$.

- (i) Find $A \times B$.
- (ii) Let R_1 be the relation from A to B consisting of ordered pairs (a, b) such that a is the capital of b . List the elements of R_1 .
- (iii) Let R_2 be the relation from A to B consisting of ordered pairs (a, b) such that a is a city of b . List the elements of R_2 .

Solution:

- (i) $A \times B = \{(k, m), (k, t), (k, c), (k, i), (j, m), (j, t), (j, c), (j, i), (p, m), (p, t), (p, c), (p, i), (s, m), (s, t), (s, c), (s, i), (l, m), (l, t), (l, c), (l, i)\}$.
- (ii) $R_1 = \{(k, m), (j, i)\}$.
- (iii) $R_2 = \{(k, m), (p, m), (s, c), (j, i)\}$

Example 2.2.3:

Let $A = \mathbb{Z}$ and a relation R on A be defined by: $x R y \Leftrightarrow x$ divides y . Determine whether the following is true or false.

- (i) $4 R 12$ _____
- (ii) $12 R 4$ _____
- (iii) $3 R 7$ _____
- (iv) $63 R 3$ _____

Definition 2.2.3 : Domain and Range of R

Let $R \subseteq A \times B$ be a relation from A to B .

The **domain of R**: $\text{Dom}(R) = \{a \in A \mid a R b \text{ for some } b \in B\}$
 = The set of all the a values of (a, b) in R .

The **range of R**: $\text{Ran}(R) = \{b \in B \mid a R b \text{ for some } a \in A\}$
 = The set of all the b values of (a, b) in R .

Example 2.2.4:

Find the domain and range of the following relations R :

- (i) $R = \{(a, a), (a, b), (b, b), (b, c), (d, c), (d, e)\}$
- (ii) $A = \{-2, -1, 0, 1, 2, 3\}$, $B = \{0, 1, 4\}$ and $x R y \Leftrightarrow y = x^2$.

Definition 2.2.4 : Matrix of a relation

Let $A = \{a_1, a_2, \dots, a_m\}$ and $B = \{b_1, b_2, \dots, b_n\}$ and let R be a relation from A to B .

The **matrix of R** is denoted $M_R = [m_{ij}]$ where $m_{ij} = \begin{matrix} 1 & \text{if } a_i R b_j \\ 0 & \text{if } a_i \not R b_j \end{matrix}$

Example 2.2.5:

Let $A = \{a, b, c\}$, $B = \{1, 2, 4, 5, 6\}$ and

let $R = \{(a, 2), (a, 5), (b, 5), (b, 6), (c, 1), (c, 4)\}$ be the relation from A to B .

$$\text{The matrix of } R = M_R = \begin{matrix} & \begin{matrix} 1 & 2 & 4 & 5 & 6 \end{matrix} \\ \begin{matrix} a \\ b \\ c \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 \end{bmatrix} \end{matrix}.$$

Some Pictorial representations of relations

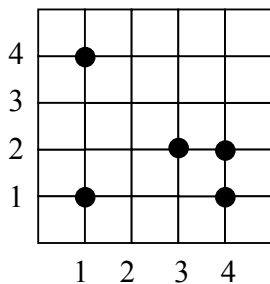
Let A and B be two nonempty sets and let R be the relation from A to B . The following are some different ways of picturing the relation R :

1. Coordinate diagram of $A \times B$
2. The matrix of R
3. The arrow diagram
4. The digraph of R

Example 2.2.6: Four different ways of picturing the relation R

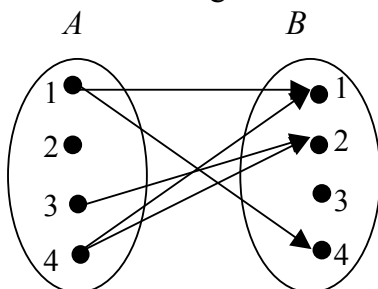
Let $A = \{1, 2, 3, 4\}$ and let $R = \{(1, 1), (1, 4), (3, 2), (4, 1), (4, 2)\}$ be the relation on A .

1. Coordinate diagram of $A \times B$
2. The matrix of R

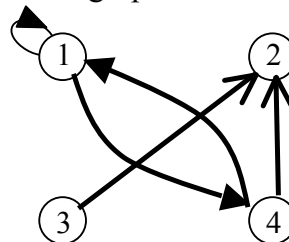


$$\begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix} \end{matrix}$$

3. The arrow diagram



4. The digraph of R



Properties of Relations

Definition 2.2.5: Reflexive, Symmetric and Transitive

Let R be a relation on a set A .

1. R is **reflexive** if $aRa \ \forall a \in A$. (i.e. every element $a \in A$ is related to itself.)
2. R is **symmetric** if $\forall a, b \in A, aRb \Rightarrow bRa$.
3. R is **transitive** if $\forall a, b, c \in A, aRb$ and $bRc \Rightarrow aRc$.

Remark:

Let R be a relation on a set A .

1. R is **not reflexive** provided $\exists a \in A$ such that $(a, a) \notin R$.
2. R on a set A is **not symmetric** provided $\exists (a, b) \in R$ but $(b, a) \notin R$.
3. R is **not transitive** provided $\exists (a, b) \in R$ and $(b, c) \in R$ but $(a, c) \notin R$.
If no such pairs (a, b) and (b, c) in R , then R is considered transitive.

Example 2.2.7:

Let $A = \{1, 2, 3, 4, 5\}$.

Is each of the following relations (i) reflexive (ii) symmetric (iii) transitive?

- (a) $R_1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (2, 4), (3, 3), (4, 2), (4, 4), (5, 5)\}$.
- (b) $R_2 = \{(1, 1), (1, 3), (2, 1), (2, 3), (3, 1), (3, 3), (5, 5)\}$.
- (c) $R_3 = \{(2, 2), (3, 3)\}$.
- (d) $R_4 = \{(a, b) \in A \times A \mid a \geq b\}$.

Solution:

- (a) R_1 is reflexive and symmetric.
 $(1, 2) \in R_1$ and $(2, 4) \in R_1$ but $(1, 4) \notin R_1$. $\therefore R_1$ is not transitive.
- (b) $2 \in A$ but $(2, 2) \notin R_2$. $\therefore R_2$ is not reflexive.
 $(2, 3) \in R_2$ but $(3, 2) \notin R_2$. $\therefore R_2$ is not symmetric.
 R_2 is transitive.
- (c) $1 \in A$ but $(1, 1) \notin R_3$. $\therefore R_3$ is not reflexive.
 R_3 is symmetric and transitive.
- (d) $a \geq a \ \forall a \in A$. So, $(a, a) \in R_4 \ \forall a \in A$. $\therefore R_4$ is reflexive.
 $(2, 1) \in R_4$ but $(1, 2) \notin R_4$. $\therefore R_4$ is not symmetric.
 $\forall a, b, c \in A, (a, b) \in R_4$ and $(b, c) \in R_4 \Rightarrow a \geq b$ and $b \geq c$.
 $\Rightarrow a \geq b \geq c \Rightarrow a \geq c \Rightarrow (a, c) \in R_4$. $\therefore R_4$ is transitive.

Definition 2.2.7: Equivalence Relation

A relation R on a set A is called an **equivalence relation** if R is reflexive, symmetric and transitive.

Example 2.2.8:

None of the relations in Example 2.2.7 is an equivalence relation.

Example 2.2.9:

Let S be the relation on the set of all real number defined by $S = \{(a, b) \in R \times R \mid a = b\}$.

Prove that S is an equivalence relation on R .

Solution:

$a = a \forall a \in R$. So, $(a, a) \in S \forall a \in R$. $\therefore S$ is reflexive. ----- (1)

$\forall a, b \in R, (a, b) \in S \Rightarrow a = b \Rightarrow b = a \Rightarrow (b, a) \in S$. $\therefore S$ is symmetric. ----- (2)

$\forall a, b, c \in R, (a, b) \in S$ and $(b, c) \in S \Rightarrow a = b$ and $b = c \Rightarrow a = b = c \Rightarrow a = c \Rightarrow (a, c) \in S$. $\therefore S$ is transitive. ----- (3)

(1), (2) and (3) $\Rightarrow S$ is an equivalence relation on R .

2.3 Functions**Definition 2.3.1 : Functions**

Let X and Y be two nonempty sets. A **function** f from X to Y is a rule that assigns a unique element of Y to each element of X .

The set X is called the **domain of f** and the set Y is the **codomain**; and we write $f: X \rightarrow Y$.

For each $x \in X$, we write $f(x) = y$ if y is assigned to x .

$f(x)$ (read as “ **f of x** ”) is called **the image of x under f** , or **the value of f at x** .

The set of all such image values is called the **image of f** , and is denoted by **$\text{Im}(f)$** .

Remark: A function from X to Y is a relation but not all relations from X to Y are functions.

Example 2.3.1:

Let $A = \{a, b, c, d\}$ and $B = \{1, 2, 3, 4, 5\}$. Which of the following relations are functions map from A to B ?

(i) $R_1 = \{(a, 2), (b, 1), (b, 3), (c, 1), (c, 4), (d, 1)\}$.

(ii) $R_2 = \{(a, 2), (b, 1), (c, 3), (d, 5)\}$.

(iii) $R_3 = \{(a, 2), (b, 1), (d, 5)\}$.

Example 2.3.2:

Let $f: \mathbb{N} \rightarrow \mathbb{Z}$ be the function defined by $f(x) = \begin{cases} 2x, & \text{if } x \text{ is odd} \\ x-1, & \text{if } x \text{ is even} \end{cases}$.

Find (i) the domain and the codomain of f ,

(ii) the image of 4, the image of 11 and the image of f .

Graph of functions

Definition 2.3.2:

Let f be function. The *graph* of f consists of all the points (x, y) where $y = f(x)$.

Example 2.3.3:

Sketch the graph of the following functions:

$$(i) \quad f(x) = \frac{x^2 - 4}{x - 2} \text{ where } -2 \leq x \leq 3, x \neq 2. \quad (ii) \quad g(x) = \begin{cases} x + 2, & x \leq 0 \\ x^2, & 0 < x \leq 2 \\ 4, & x > 2 \end{cases}$$

One-to-one, onto and invertible functions

Definition 2.3.3: One-to-one and onto

A function $f: X \rightarrow Y$ is said to be **one-to-one** (written **1-1**) if

$$\forall x_1, x_2 \in X, \quad f(x_1) = f(x_2) \Rightarrow x_1 = x_2.$$

Or equivalent, $\forall x_1, x_2 \in X, \quad x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2)$.

A function $f: X \rightarrow Y$ is said to be **onto** if $\forall y \in Y, \exists x \in X$ such that $f(x) = y$.

In other words, $\text{Im}(f) = Y$ (i.e. The image of f = the codomain of f), and we say f is a function **from X onto Y** , or **f maps X onto Y** .

A function $f: X \rightarrow Y$ is called **one-to-one correspondence** if it is one-to-one and onto.

Remark: Some texts use the term **injective** for a 1-1 function, **surjective** for an onto function and **bijective** for an one-to-one correspondence function.

Example 2.3.4:

Is each of the following function (i) one-to-one (ii) onto (iii) one-to-one correspondence?

(a) $f: \mathbb{R} \rightarrow \mathbb{R}$ be the function defined by $f(x) = x + 1$.

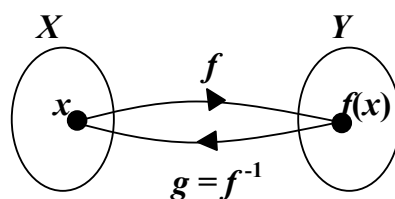
(b) $g: \mathbb{N} \rightarrow \mathbb{R}$ be the function defined by $g(x) = \sqrt{x}$.

(c) $h: \mathbb{R} \rightarrow \mathbb{R}$ be the function defined by $h(x) = \sin x$.

Definition 2.3.4: Inverse Functions

Let $f: X \rightarrow Y$ be a function. We say f is **invertible** if there exists a function $g: Y \rightarrow X$ such that $\forall x \in X$ and $\forall y \in Y, \quad f(x) = y \Leftrightarrow g(y) = x$.

In this case, we call such a function g an **inverse of f** and we denoted g by f^{-1} .



Remark: f^{-1} (read as *f inverse*) does not mean $\frac{1}{f}$.

Theorem 2.3.1:

A function $f: X \rightarrow Y$ is invertible if and only if f is one-to-one correspondence.

Example 2.3.5:

Which of the following functions are invertible?

- (a) $f: \mathbb{R} \rightarrow \mathbb{R}$ be the function defined by $f(x) = x^2$.
- (b) $g: \{x \mid x \geq 0\} \rightarrow \{x \mid x \geq 0\}$ be the function defined by $g(x) = x^2$.
- (b) $h: \mathbb{N} \rightarrow \mathbb{N}$ be the function defined by $h(x) = 2x$.
- (c) $k: \mathbb{R} \rightarrow \mathbb{R}$ be the function defined by $k(x) = \sqrt[3]{x}$.

Composite Functions

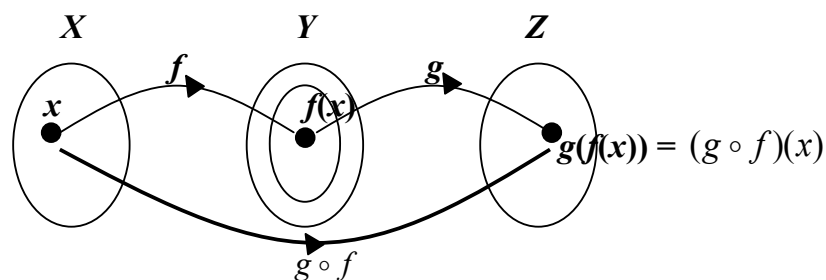
Definition 2.3.5: Composite Functions

Let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ be functions with the property that $\text{Im}(f) \subseteq \text{Dom}(g)$.

Define a new function $g \circ f: X \rightarrow Z$ as follows:

$$(g \circ f)(x) = g(f(x)) \quad \forall x \in X.$$

$g \circ f$ is called **the composition of f and g** . (We put the f first when we say “the composition of f and g ” because an element x is first acted upon by f and then by g .)



Remark:

- In general, $g \circ f \neq f \circ g$.
- If $f: X \rightarrow Y$ is any function, then $f \circ 1_X = f = 1_Y \circ f$ where 1_X is the **identity function on X** , that is $1_X(x) = x \quad \forall x \in X$.

Example 2.3.6:

Let $f: \{a, b, c\} \rightarrow \{1, 2, 3\}$ be the function $\{(a, 2), (b, 1), (c, 2)\}$ and let $g: \{1, 2, 3\} \rightarrow \{a, c, d\}$ be the function $\{(1, c), (2, d), (3, a)\}$.

Find (i) $g \circ f$, if exists (ii) $f \circ g$, if exists.

Chapter 3

Complex Numbers

3.1 Introduction

To have solutions for all equations $ax = b$ where a, b are integers, $a \neq 0$, the integers were extended to the rational numbers. To have a solution for $x^2 = 2$, we went outside the rational numbers to the irrational numbers. There is no real number x such that $x^2 = -1$ since the square of any nonzero real number must be positive. To have a solution for $x^2 = -1$, mathematicians introduced the complex number i with the property $i^2 = -1$. We also use $\sqrt{-1}$ to denote i . Thus the equation $x^2 + 1 = 0$ would have two roots $\pm i$. Also, the square root of a negative number, $-b^2$, is written in the form bi , where b is a real number. Furthermore, solutions of the equation $(x - a)^2 = -b^2$ (a, b both real numbers) are given by $x - a = \pm bi$, that is $x = a \pm bi$.

Complex numbers made their appearance in the mathematical scene in The Great Art, or in the rules of Algebra (1545) by the Italian mathematician and physician Gerolamo Cardano (1501 - 1576). Cardano studied the quadratic problem of dividing 10 into two parts such that their product is 40 and found the solution $5 + \sqrt{-15}$ and $5 - \sqrt{-15}$ by standard technique. About 27 years later the engineer Raphael Bombelli (1526 - 1572) published an algebra text in which he dealt systematically with complex numbers.

Definition 3.1.1 *A complex number z is any expression of the form*

$$z = a + bi,$$

where a and b are real numbers and $i = \sqrt{-1}$. The scalar a is called the real part of z and b is called the imaginary part of z .

Notation 1 If $z = a + bi$, $a, b \in \mathbb{R}$, then we use $\text{Re}(z)$ to denote the number a and $\text{Im}(z)$ to denote the number b . $\mathbb{C} = \{a + bi : a, b \in \mathbb{R}\}$ denotes the set of all complex numbers.

Remark 3.1.1 If $b = 0$, we write $a + 0i$ as a . We identify $a + 0i$ as the real number a . If $a = 0$, we write $0 + bi$ as bi ; and bi is called a pure imaginary number.

Every complex number is the sum of a real number and a pure imaginary number.

Example 3.1.1 If $z = 3 + 4i$, then $\text{Re}(z) = 3$, $\text{Im}(z) = 4$ (3 is the real part of z and 4 is the imaginary part of z).

Definition 3.1.2 (i) A complex number $a + bi$ ($a, b \in \mathbb{R}$) where $b \neq 0$ is called imaginary; i.e. a complex number that is not real is imaginary.

(ii) Two complex numbers $a + bi$ and $c + di$ ($a, b, c, d \in \mathbb{R}$) are said to be equal if $a = c$ and $b = d$.

3.2 Operations on complex numbers

Let $z_1 = a + bi$, $z_2 = c + di$ ($a, b, c, d \in \mathbb{R}$)

1. Addition: $z_1 + z_2 = (a + c) + (b + d)i$
2. Subtraction: $z_1 - z_2 = (a - c) + (b - d)i$
3. Multiplication: $z_1 z_2 = (a + bi)(c + di) = (ac - bd) + (ad + bc)i$
4. Division: For $z_2 \neq 0$,

$$\frac{z_1}{z_2} = \frac{a + bi}{c + di} = \frac{a + bi}{c + di} \cdot \frac{c - di}{c - di} = \frac{ac + bd}{c^2 + d^2} + \frac{bc - ad}{c^2 + d^2}i$$

Here 0 stands for $0 + 0i$. For $z = a + bi \in \mathbb{C}$, let $\bar{z} = a - bi$. Then $\bar{z}$ is called the conjugate of z . The conjugate of z is also sometimes denoted by z^* . Note that

$$z\bar{z} = (a + bi)(a - bi) = a^2 + b^2$$

Remark 3.2.1 1. $z_1 + z_2 = z_2 + z_1 \quad \forall z_1, z_2 \in \mathbb{C}$ (commutative law)

2. $z_1 + (z_2 + z_3) = (z_1 + z_2) + z_3 \quad \forall z_1, z_2, z_3 \in \mathbb{C} \quad (\text{associative law})$
3. $z_1 z_2 = z_2 z_1 \quad \forall z_1, z_2 \in \mathbb{C} \quad (\text{commutative law})$
4. $(z_1 z_2) z_3 = z_1 (z_2 z_3) \quad \forall z_1, z_2, z_3 \in \mathbb{C} \quad (\text{associative law})$
5. $(z_1 + z_2) z_3 = z_1 z_3 + z_2 z_3 \quad \forall z_1, z_2, z_3 \in \mathbb{C} \quad (\text{distributive law})$
6. $z = \bar{z} \Leftrightarrow z \text{ is a real number}$
7. $\frac{z_1}{w_1} \cdot \frac{z_2}{w_2} = \frac{z_1 z_2}{w_1 w_2}$ if $z_1, z_2, w_1, w_2 \in \mathbb{C}, w_1, w_2 \neq 0$.

Example 3.2.1 1. Factorize

(a) $x^2 + 1$

(b) $(x + p)^2 + q^2$

2. Solve the equation $x^2 - 2x + 8 = 0$.

3. Let $z_1 = 2 + 4i$, $z_2 = -1 + 3i$. Compute $z_1 + z_2$, $z_1 - z_2$, $z_1 z_2$, $z_2 \bar{z}_2$, $\frac{z_1}{z_2}$.

SOLUTION.

1. (a) $x^2 + 1 = x^2 - i^2 = (x + i)(x - i)$

(b) $(x + p)^2 + q^2 = (x + p)^2 - i^2 q^2 = (x + p + qi)(x + p - qi)$

2. $x = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(8)}}{2} = \frac{2 \pm \sqrt{-28}}{2} = \frac{2 \pm \sqrt{28}i}{2} = 1 \pm \sqrt{7}i$.

3. $z_1 + z_2 = (2 - 1) + (4 + 3)i = 1 + 7i$

$z_1 - z_2 = (2 + 1) + (4 - 3)i = 3 + i$

$z_1 z_2 = (2 + 4i)(-1 + 3i) = -2 + 6i - 4i + 12i^2 = -2 + 2i - 12 = -14 + 2i$

$\bar{z}_2 = -1 - 3i$, $z_2 \bar{z}_2 = (-1)^2 + (-3)^2 = 10$

$\frac{z_1}{z_2} = \frac{z_1 \bar{z}_2}{z_2 \bar{z}_2} = \frac{(2 + 4i)(-1 - 3i)}{10} = \frac{-2 - 6i - 4i - 12i^2}{10} = \frac{-2 - 10i + 12}{10} = \frac{10 - 10i}{10} = 1 - i$

Theorem 3.2.1 For any complex numbers z and w ,

(i) $\overline{z + w} = \bar{z} + \bar{w}$

(ii) $\overline{zw} = \bar{z} \bar{w}$

(iii) $\overline{\left(\frac{z}{w}\right)} = \frac{\bar{z}}{\bar{w}}$ whenever $w \neq 0$.

PROOF.

Let $z = a + bi$ and $w = c + di$. Then $\bar{z} = a - bi$ and $\bar{w} = c - di$.

$$\begin{aligned} \text{(i)} \quad z + w &= (a + c) + (b + d)i \\ \overline{z + w} &= (a + c) - (b + d)i = (a - bi) + (c - di) = \bar{z} + \bar{w} \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad zw &= (ac - bd) + (ad + bc)i \\ \overline{zw} &= (ac - bd) - (ad + bc)i \\ \bar{z} \bar{w} &= (a - bi)(c - di) = (ac - bd) - (ad + bc)i = \overline{zw} \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad \frac{z}{w} &= \frac{a+bi}{c+di} \cdot \frac{c-di}{c-di} = \frac{(ac+bd)+(bc-ad)i}{c^2+d^2} \\ \overline{\left(\frac{z}{w}\right)} &= \frac{(ac+bd)-(bc-ad)i}{c^2+d^2} \\ \frac{\bar{z}}{\bar{w}} &= \frac{a-bi}{c-di} = \frac{a-bi}{c-di} \cdot \frac{c+di}{c+di} = \frac{(ac+bd)+(ad-bc)i}{c^2+d^2} = \overline{\left(\frac{z}{w}\right)} \end{aligned}$$

Remark 3.2.2 Using mathematical induction, one can show that

$$\text{(i)} \quad \overline{(z_1 + z_2 + \dots + z_n)} = \bar{z}_1 + \bar{z}_2 + \dots + \bar{z}_n$$

$$\text{(ii)} \quad \overline{z_1 z_2 \dots z_n} = \bar{z}_1 \cdot \bar{z}_2 \dots \bar{z}_n$$

Example 3.2.2 Verify Theorem 3.2.1 for $z_1 = 2 - 7i, w = 3 + i$.

SOLUTION.

$$\bar{z} = 2 + 7i, \bar{w} = 3 - i, z + w = 5 - 6i.$$

$$\begin{aligned} \text{(i)} \quad \overline{z + w} &= 5 + 6i. \\ \therefore \bar{z} + \bar{w} &= (2 + 3) + (7 - 1)i = 5 + 6i = \overline{z + w}. \end{aligned}$$

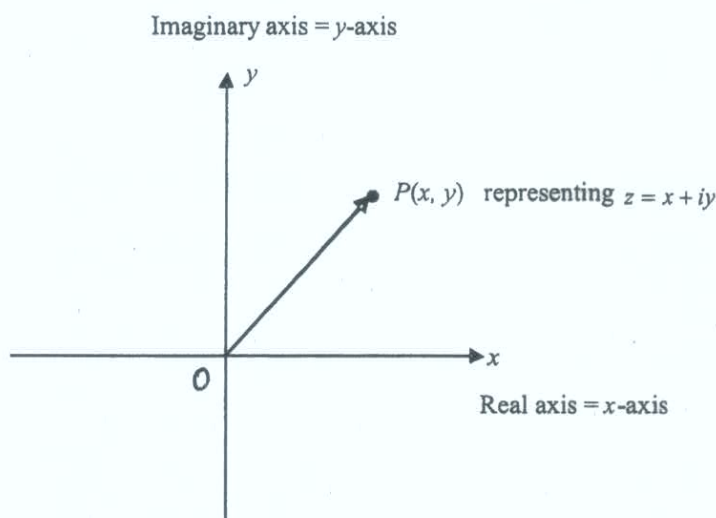
$$\begin{aligned} \text{(ii)} \quad \overline{zw} &= \overline{13 - 19i} = 13 + 19i \\ \bar{z} \bar{w} &= (2 + 7i)(3 - i) = 6 + 7 + (-2 + 21)i = 13 + 19i = \overline{zw} \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad \frac{z}{w} &= \frac{2-7i}{3+i} \cdot \frac{3-i}{3-i} = \frac{6-7-(2+21)i}{3^2+1^2} = \frac{1}{10}(-1 - 23i) \\ \overline{\left(\frac{z}{w}\right)} &= \frac{1}{10}(-1 + 23i) \\ \frac{\bar{z}}{\bar{w}} &= \frac{2+7i}{3-i} \cdot \frac{3+i}{3+i} = \frac{6-7+(2+21)i}{3^2+1^2} = \frac{1}{10}(-1 + 23i) = \overline{\left(\frac{z}{w}\right)} \end{aligned}$$

3.3 The Argand Plane

Let $z = x + yi$ ($x, y \in \mathbb{R}$) and P be the point (x, y) in the xy -plane. Then z can be represented by

1. the point P , or
2. the vector $\overrightarrow{OP}$.



As the xy -plane is used to represent complex numbers, it is called the complex plane or Argand plane. As the points $(x, 0)$ correspond to real numbers $x + 0i$, the x -axis is called the real axis. Similarly, the y -axis is called the imaginary axis because points on it have the form $0 + iy$ for y real.

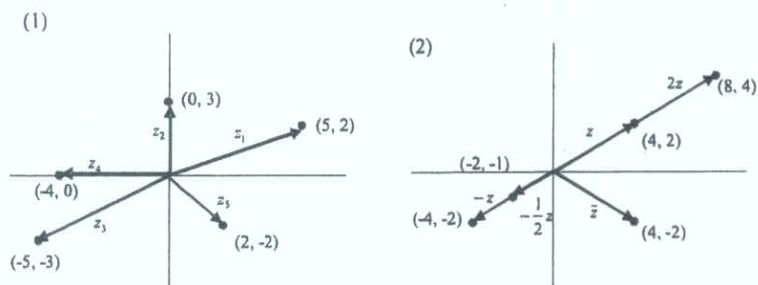
Historically, the geometric representation of complex numbers in the xy -plane was due to Gauss (1777 – 1855) and two lesser known mathematicians Jean Robert Argand (1786 – 1822) and Caspar Wessel (1745 – 1818).

Example 3.3.1 1. Represent the following complex numbers in the Argand plane:

$$z_1 = 5 + 2i, z_2 = 3i, z_3 = -5 - 3i, z_4 = -4, z_5 = 2 - 2i.$$

2. Let $z = 4 + 2i$. Represent $z, \bar{z}, 2z, -z$ and $\frac{-z}{2}$ in the Argand plane.

SOLUTION.



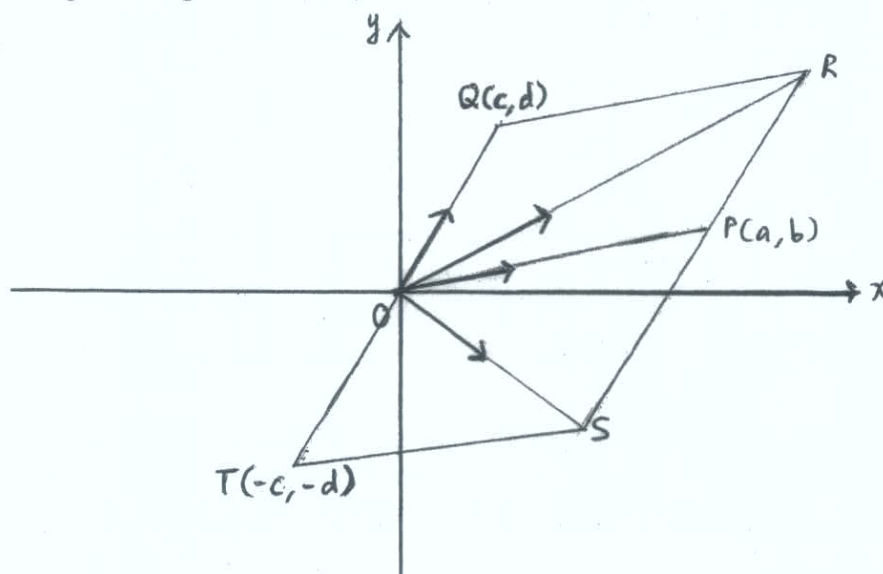
3.3.1 Representation of Addition and Subtraction

Let $z_1 = a + bi$, $z_2 = c + di$ ($a, b, c, d \in \mathbb{R}$). Then $z_1 + z_2 = (a + c) + (b + d)i$.

z_1 is represented by the point $P(a, b)$.

z_2 is represented by the point $Q(c, d)$.

Draw the parallelograms $OPRQ$ and $OPST$ as follows:



Then R is the point $(a + c, b + d)$ and hence R represents $z_1 + z_2$. Note that

$$\vec{OR} = \vec{OP} + \vec{PR} = \vec{OP} + \vec{OQ}.$$

In the argand diagram, addition of complex numbers corresponds to addition of vectors.

S is the point $(a - c, b - d)$ and hence S represents $z_1 - z_2$. Note that

$$\overrightarrow{OS} = \overrightarrow{OP} + \overrightarrow{PS} = \overrightarrow{OP} + \overrightarrow{OT} = \overrightarrow{OP} - \overrightarrow{OQ}.$$

In the argand diagram, subtraction of complex numbers corresponds to subtraction of vectors.

3.4 Polar forms of complex numbers

A complex number written in the form $z = x + yi$, $x, y \in \mathbb{R}$, is said to be represented in Cartesian form.

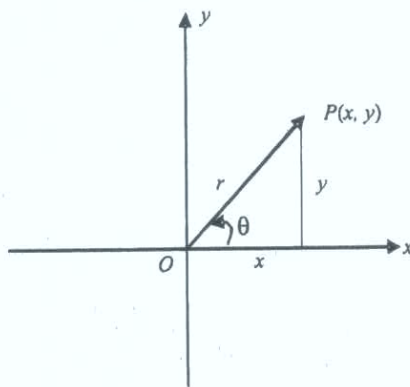


Figure 3.1: A representation of the complex number $z = x + yi$

For $z = x + yi \neq 0$, Figure (a) indicates that if θ is an angle from the positive x -axis to the vector representation of z , r is the distance of the point $P(x, y)$ from the origin. Then

- (i) $r = \sqrt{x^2 + y^2}$, $\tan \theta = \frac{y}{x}$
- (ii) $x = r \cos \theta$, $y = r \sin \theta$
- (iii) $z = r(\cos \theta + i \sin \theta)$

Form z in (iii) is called the polar form of z . The length of vector $\overrightarrow{OP}$, that is r , is called the modulus or absolute value of z and is denoted by $|z|$. $\therefore |z| = r = \sqrt{x^2 + y^2}$.

The angle θ is called an argument of z and is denoted by $\arg(z)$. Of particular importance is the restricted value $0 \leq \theta < 2\pi$. The principal argument

of z is denoted by $\text{Arg}(z)$. We usually use this principal argument of z and refer to the corresponding form as the polar form of z .

Example 3.4.1 Find the principal argument and the polar form of $z = 1 - i$.

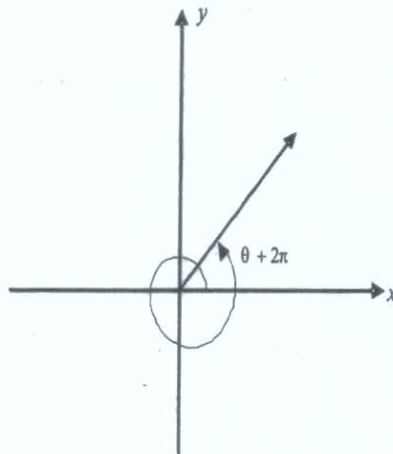
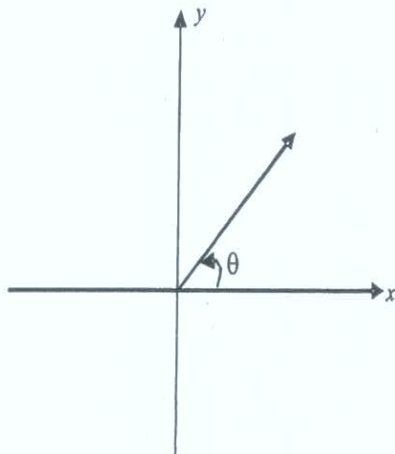
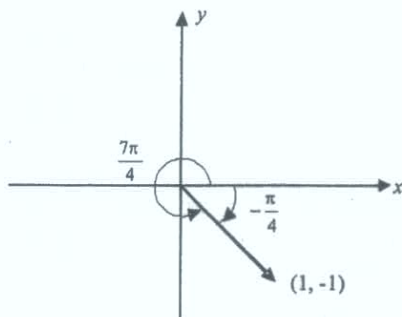
SOLUTION.

$\tan \theta = \frac{y}{x} = -1$. $(1, -1)$ is in the fourth quadrant.

$\therefore$ the principal argument, $\text{Arg}(z) = \frac{7\pi}{4}$.

$$r = \sqrt{1^2 + (-1)^2} = \sqrt{2},$$

$z = \sqrt{2} \left(\cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4} \right)$ is the polar form of z .



Remark 3.4.1 (i) The value of $\arg(z)$ is not unique. If θ is a value of $\arg(z)$, then $\theta + 2n\pi$, $n = 0, \pm 1, \pm 2, \dots$, is also a value of $\arg(z)$.

(ii) Let $z = x + yi, z \neq 0$.

If (x, y) is in the 1st quadrant, then $0 \leq \text{Arg}(z) \leq \frac{\pi}{2}$.

If (x, y) is in the 2nd quadrant, then $\frac{\pi}{2} \leq \text{Arg}(z) \leq \pi$.

If (x, y) is in the 3rd quadrant, then $\pi \leq \text{Arg}(z) \leq \frac{3\pi}{2}$.

If (x, y) is in the 4th quadrant, then $\frac{3\pi}{2} \leq \text{Arg}(z) \leq 2\pi$.

Example 3.4.2 Find the modulus and principal arguments for the following complex numbers:

(i) $z_1 = \sqrt{3} + i$,

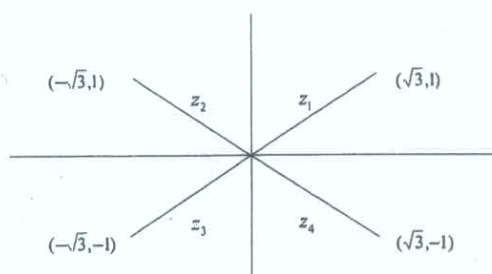
(ii) $z_2 = -\sqrt{3} + i$,

(iii) $z_3 = -\sqrt{3} - i$,

(iv) $z_4 = \sqrt{3} - i$,

SOLUTION.

$$z_1 = \sqrt{(\sqrt{3})^2 + 1^2} = 2 = |z_2| = |z_3| = |z_4|.$$



(i) Let $\theta_1 = \text{Arg}(z_1)$. Then $\tan \theta_1 = \frac{1}{\sqrt{3}}$ where $0 \leq \theta_1 \leq \frac{\pi}{2}$. $\therefore \text{Arg}(z_1) = \frac{\pi}{6}$.
(The polar form of $z_1 = 2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)$).

(ii) Let $\theta_2 = \text{Arg}(z_2)$. Then $\tan \theta_2 = \frac{1}{-\sqrt{3}} = -\frac{1}{\sqrt{3}}$ where $\frac{\pi}{2} \leq \theta_2 \leq \pi$.
 $\therefore \text{Arg}(z_2) = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$.
(The polar form of $z_2 = 2 \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right)$).

(iii) Let $\theta_3 = \text{Arg}(z_3)$. Then $\tan \theta_3 = \frac{-1}{-\sqrt{3}} = \frac{1}{\sqrt{3}}$ where $\pi \leq \theta_3 \leq \frac{3\pi}{2}$.
 $\therefore \text{Arg}(z_3) = \pi + \frac{\pi}{6} = \frac{7\pi}{6}$.
(The polar form of $z_3 = 2 \left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6} \right)$).

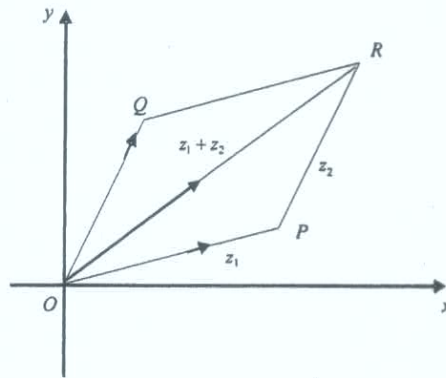
- (iv) Let $\theta_4 = \text{Arg}(z_4)$. Then $\tan \theta_4 = \frac{-1}{\sqrt{3}} = -\frac{1}{\sqrt{3}}$ where $\frac{3\pi}{2} \leq \theta_4 \leq 2\pi$.
 $\therefore \text{Arg}(z_4) = 2\pi - \frac{\pi}{6} = \frac{11\pi}{6}$.
 (The polar form of $z_3 = 2 \left(\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6} \right)$).

3.4.1 Triangular Inequalities

Theorem 3.4.1 For any complex numbers z_1 and z_2 ,

$$|z_1 + z_2| \leq |z_1| + |z_2|.$$

PROOF.



From $\triangle OPR$, we have $OR \leq OP + PR$, that is $|z_1 + z_2| \leq |z_1| + |z_2|$.

3.4.2 Multiplication in polar forms

Let $z_1 = r_1 (\cos \theta_1 + i \sin \theta_1)$, $z_2 = r_2 (\cos \theta_2 + i \sin \theta_2)$. Then

$$\begin{aligned} z_1 z_2 &= r_1 r_2 (\cos \theta_1 + i \sin \theta_1) (\cos \theta_2 + i \sin \theta_2) \\ &= r_1 r_2 [(\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2) + i (\sin \theta_1 \cos \theta_2 + \sin \theta_2 \cos \theta_1)] \\ \therefore z_1 z_2 &= r_1 r_2 [\cos (\theta_1 + \theta_2) + i \sin (\theta_1 + \theta_2)] \end{aligned}$$

$$|z_1 z_2| = r_1 r_2 = |z_1| |z_2|, \quad \text{Arg}(z_1) + \text{Arg}(z_2) \text{ is an argument of } z_1 z_2.$$

From the above identity, we see that when two complex numbers are multiplied, the moduli are multiplied and the arguments are added. Notice, however, that the principal argument of a product may not be the sum of the principal arguments of the factors; the sum of the principal arguments may be $\geq 2\pi$ and have to be adjusted to lie between 0 and 2π .

Similarly it can be shown that if $z_2 \neq 0$, then

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} [\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)]$$

and hence $\left| \frac{z_1}{z_2} \right| = \frac{r_1}{r_2} = \frac{|z_1|}{|z_2|}$ and $\text{Arg}(z_1) - \text{Arg}(z_2)$ is an argument of $\frac{z_1}{z_2}$.

Example 3.4.3 Let $z = 2(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})$, $w = \sqrt{3}(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4})$. Find

(i) zw ;

(ii) $\frac{z}{w}$.

SOLUTION.

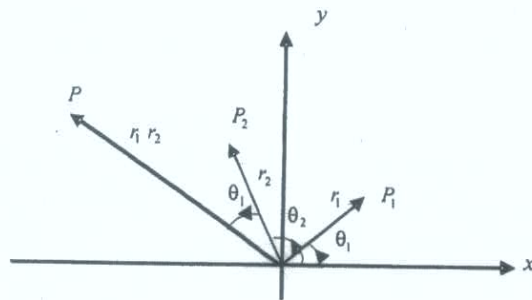
$$\begin{aligned} \text{(i)} \quad zw &= 2\sqrt{3} \left[\cos \left(\frac{\pi}{3} + \frac{3\pi}{4} \right) + i \sin \left(\frac{\pi}{3} + \frac{3\pi}{4} \right) \right] \\ &= 2\sqrt{3} \left[\cos \left(\frac{13\pi}{12} \right) + i \sin \left(\frac{13\pi}{12} \right) \right] \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \frac{z}{w} &= \frac{2}{\sqrt{3}} \left[\cos \left(\frac{\pi}{3} - \frac{3\pi}{4} \right) + i \sin \left(\frac{\pi}{3} - \frac{3\pi}{4} \right) \right] \\ &= \frac{2}{\sqrt{3}} \left[\cos \left(-\frac{5\pi}{12} \right) + i \sin \left(-\frac{5\pi}{12} \right) \right] \\ &= \frac{2}{\sqrt{3}} \left[\cos \left(\frac{5\pi}{12} \right) - i \sin \left(\frac{5\pi}{12} \right) \right] \end{aligned}$$

3.4.3 Geometric interpretation

Let $z_1 = r_1(\cos \theta_1 + i \sin \theta_1)$, $z_2 = r_2(\cos \theta_2 + i \sin \theta_2)$ be represented by vectors $\overrightarrow{OP_1}$ and $\overrightarrow{OP_2}$ respectively.

The diagram shows that the vector $\overrightarrow{OP}$ representing $z_1 z_2$ is obtained by rotating the vector $\overrightarrow{OP_2}$ through the angle θ_1 in the counterclockwise direction and by multiplying the length r_2 of $\overrightarrow{OP_2}$ with the length r_1 of $\overrightarrow{OP_1}$.



3.4.4 Properties of Modulus

Theorem 3.4.2 Let $z, z_1, z_2, \dots, z_n$ be complex numbers. Then

- (i) $|z| \geq 0$ and $|z| = 0$ if and only if $z = 0$;
- (ii) $|z| = |-z| = |\bar{z}|$;
- (iii) $z\bar{z} = |z|^2 = (\operatorname{Re}(z))^2 + (\operatorname{Im}(z))^2$;
- (iv) $|z_1 z_2 \dots z_n| = |z_1| |z_2| \dots |z_n|$;
- (v) $\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$ for nonzero z_2 ;
- (vi) $|z_1 + z_2 + \dots + z_n| \leq |z_1| + |z_2| + \dots + |z_n|$;
- (vii) $|z_1 \pm z_2| \geq ||z_1| - |z_2||$;
- (viii) If $z \neq 0$, then $z^{-1} = \frac{\bar{z}}{|z|^2}$.

3.5 de Moivre's Theorem

For any $\theta \in \mathbb{R}$ and integer n ,

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta \quad (*)$$

PROOF.

Case 1: $n = 0$.

LHS = 1, RHS = $\cos 0 + i \sin 0 = 1$. $\therefore (*)$ is true.

Case 2: $n \geq 1$.

When $n = 1$, LHS = $\cos \theta + i \sin \theta$ = RHS.

Assume that (*) is true for $n = k \geq 1$, i.e.

$$(\cos \theta + i \sin \theta)^k = \cos k\theta + i \sin k\theta. \quad (**)$$

Then

$$\begin{aligned} & (\cos \theta + i \sin \theta)^{k+1} \\ &= (\cos \theta + i \sin \theta)^k (\cos \theta + i \sin \theta) \\ &= (\cos k\theta + i \sin k\theta)(\cos \theta + i \sin \theta) \quad (\text{by } (**)) \\ &= (\cos k\theta \cos \theta - \sin k\theta \sin \theta) + i(\sin k\theta \cos \theta + \cos k\theta \sin \theta) \\ &= \cos(k\theta + \theta) + i \sin(k\theta + \theta) \\ &= \cos(k+1)\theta + i \sin(k+1)\theta \end{aligned}$$

$\therefore$ (*) is true for $n = k \Rightarrow$ (*) is true for $n = k+1$. It follows by induction that (*) is true for all positive integer n .

Case 3: $n < 0$. Let $n = -m$. Then $m > 0$ and

$$\begin{aligned} (\cos \theta + i \sin \theta)^n &= (\cos \theta + i \sin \theta)^{-m} \\ &= \frac{1}{(\cos \theta + i \sin \theta)^m} \\ &= \frac{1}{(\cos(m\theta) + i \sin(m\theta))} \quad (\text{from case 2}) \\ &= \frac{1}{(\cos(m\theta) + i \sin(m\theta))} \cdot \frac{(\cos(m\theta) - i \sin(m\theta))}{(\cos(m\theta) - i \sin(m\theta))} \\ &= (\cos(m\theta) - i \sin(m\theta)) \\ &= (\cos(-m\theta) + i \sin(-m\theta)) \\ &= (\cos(n\theta) + i \sin(n\theta)). \end{aligned}$$

Remark 3.5.1 Let $z = r(\cos \theta + i \sin \theta)$ and n a positive integer. Then by de Moivre's Theorem,

$$z^n = [r(\cos \theta + i \sin \theta)]^n = r^n (\cos \theta + i \sin \theta)^n = r^n (\cos n\theta + i \sin n\theta)$$

$$z^{-n} = [r(\cos \theta + i \sin \theta)]^{-n} = r^{-n} (\cos \theta + i \sin \theta)^{-n} = \frac{1}{r^n} (\cos(-n\theta) + i \sin(-n\theta))$$

$$\therefore \frac{1}{z^n} = \frac{1}{r^n} (\cos(n\theta) - i \sin(n\theta))$$

Example 3.5.1 Given $z = 2(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3})$, find z^{12} .

SOLUTION.

$$z^{12} = 2^{12}(\cos(12 \times \frac{2\pi}{3}) + i \sin(12 \times \frac{2\pi}{3})) = 2^{12}(\cos 8\pi + i \sin 8\pi) = 2^{12}$$

($\because \cos 8\pi = 1, \sin 8\pi = 0$)

Example 3.5.2 Compute $\left(\frac{1+\sqrt{3}i}{1-\sqrt{3}i}\right)^{10}$.

SOLUTION.

$$z = \frac{1+\sqrt{3}i}{1-\sqrt{3}i} = \frac{(1+\sqrt{3}i)(1+\sqrt{3}i)}{(1-\sqrt{3}i)(1+\sqrt{3}i)} = \frac{(1+\sqrt{3}i)^2}{4}.$$

$$\therefore z^{10} = \frac{(1+\sqrt{3}i)^{20}}{4^{10}}.$$

Let $w = 1 + \sqrt{3}i$. Then $|w| = \sqrt{1+3} = 2$, $\tan \frac{\pi}{3} = \sqrt{3}$.

$$\therefore w = 2(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})$$

$$\begin{aligned} z^{10} &= \frac{[2(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})]^{20}}{4^{10}} \\ &= \cos(20 \times \frac{\pi}{3}) + i \sin(20 \times \frac{\pi}{3}) \\ &= \cos(6\pi + \frac{2}{3}\pi) + i \sin(6\pi + \frac{2}{3}\pi) \\ &= \cos \frac{2}{3}\pi + i \sin \frac{2}{3}\pi \\ &= -\frac{1}{2} + \frac{\sqrt{3}}{2}i. \end{aligned}$$

3.5.1 Application of de Moivre's Theorem

Example 3.5.3 (i) Express $\cos 4\theta$ in terms of powers of $\cos \theta$.

(ii) Express $\sin 4\theta$ in terms of powers of $\sin$ and $\cos \theta$.

SOLUTION.

Let $k = \cos \theta$ and $s = \sin \theta$. Then

$$\begin{aligned} \cos 4\theta + i \sin 4\theta &= (\cos \theta + i \sin \theta)^4 \\ &= (k + si)^4 \\ &= k^4 + 4k^3(si) + 6k^2(si)^2 + 4k(si)^3 + (si)^4 \\ &= k^4 + 4k^3si - 6k^2s^2 - 4ks^3i + s^4 \\ &= (k^4 - 6k^2s^2 + s^4) + i(4k^3s - 4ks^3) \end{aligned}$$

Comparing the real and the imaginary parts,

(i)

$$\begin{aligned}\cos 4\theta &= k^4 - 6k^2s^2 + s^4 \\ &= k^4 - 6k^2(1 - k^2) + (1 - k^2)^2 \\ &= 8k^4 - 8k^2 + 1 \\ &= 8\cos^4\theta - 8\cos^2\theta + 1.\end{aligned}$$

(ii)

$$\begin{aligned}\sin 4\theta &= 4k^3s - 4ks^3 \\ &= 4\cos^3\theta \sin\theta - 4\cos\theta \sin^3\theta.\end{aligned}$$

Example 3.5.4 Let $z = \cos\theta + i\sin\theta$ and n a positive integer. Show that

(i) $2\cos n\theta = z^n + z^{-n}$;

(ii) $2i\sin n\theta = z^n - z^{-n}$.

Using (i) and (ii), express $\sin^4\theta$ in terms of multiples of θ .

SOLUTION.

By de Moivre's theorem,

$$z^n = (\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta \quad (3.1)$$

$$z^{-n} = (\cos\theta + i\sin\theta)^{-n} = \cos n\theta - i\sin n\theta \quad (3.2)$$

$$(3.1) + (3.2) : z^n + z^{-n} = 2\cos n\theta$$

$$(3.1) - (3.2) : z^n - z^{-n} = 2i\sin n\theta$$

When $n = 1$,

$$\sin\theta = \frac{1}{2i}(z - z^{-1})$$

$$\begin{aligned}
\therefore \sin 4\theta &= \left(\frac{1}{2i}\right)^4 (z - z^{-1})^4 \\
&= \frac{1}{16}(z^4 - 4z^2 + 6 - 4z^{-2} + z^{-4}) \\
&= \frac{1}{16}[(z^4 + z^{-4}) - 4(z^2 + z^{-2}) + 6] \\
&= \frac{1}{16}[2\cos 4\theta - 8\cos 2\theta + 6] \\
&= \frac{1}{8}[\cos 4\theta - 4\cos 2\theta + 3].
\end{aligned}$$

3.6 Exponential form of complex numbers

By assuming that the infinite series expansion

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

holds when $x = i\theta$, $\theta \in \mathbb{R}$, we can arrive at

$$\begin{aligned}
e^{i\theta} &= 1 + i\theta + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \frac{(i\theta)^4}{4!} + \frac{(i\theta)^5}{5!} + \dots \\
&= \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \dots\right) + i\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \dots\right)
\end{aligned}$$

that is,

$$e^{i\theta} = \cos \theta + i \sin \theta.$$

This is also known as Euler's formula. In general, for $z = x + yi$ we have

$$e^z = e^{x+iy} = e^x e^{iy} = e^x (\cos y + i \sin y), \quad x, y \in \mathbb{R}.$$

In the special case where $y = 0$, this reduces to e^x .

By de Moivre's theorem,

$$e^{-i\theta} = \cos(-\theta) + i \sin(-\theta) = \cos \theta - i \sin \theta$$

$$e^{i\theta} + e^{-i\theta} = 2 \cos \theta; \quad e^{i\theta} - e^{-i\theta} = 2i \sin \theta$$

$$\therefore \boxed{\cos \theta = \frac{1}{2} (e^{i\theta} + e^{-i\theta}), \sin \theta = \frac{1}{2i} (e^{i\theta} - e^{-i\theta})}$$

Also $(e^{i\theta})^n = e^{in\theta}$, $n \in \mathbb{Z}$. Note that $e^{i\pi} = \cos \pi + i \sin \pi = -1$. Every complex number $z = r(\cos \theta + i \sin \theta)$ can be written in exponential form as follows

$$z = re^{i\theta}.$$

Example 3.6.1 Express $z = 2i$ in exponential form.

SOLUTION.

$$|z| = |2i| = 2, \operatorname{Arg}(z) = \frac{\pi}{2}. \text{ Thus } z = 2e^{i\frac{\pi}{2}}.$$

Example 3.6.2 Express $z = 2 + 2\sqrt{3}i$ in exponential form.

SOLUTION.

$$|z| = \sqrt{2^2 + (2\sqrt{3})^2} = \sqrt{4 + 12} = 4. \quad (2, 2\sqrt{3}) \text{ is in the first quadrant,}$$

$$\theta = \tan^{-1} \frac{2\sqrt{3}}{2} = \tan^{-1} \sqrt{3} = \frac{\pi}{3}. \therefore z = 4e^{i\frac{\pi}{3}}.$$

Definition 3.6.1 A complex number z is called an n^{th} root of another complex number w if $z^n = w$. (Here n is a positive integer)

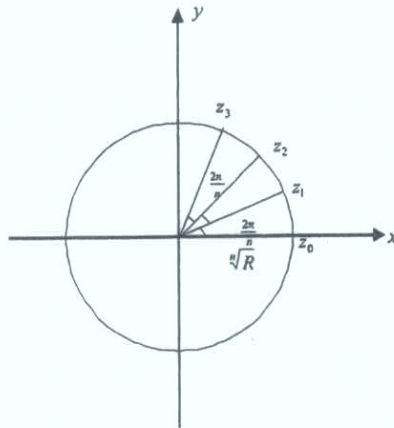
Example 3.6.3 $1, -1, i, -i$ are the fourth roots of unity; they are solutions of the equation $z^4 = 1$.

Let $w = R(\cos \alpha + i \sin \alpha)$ and $z = r(\cos \theta + i \sin \theta)$. Let n be a positive integer. Suppose that $z^n = w$. Then

$$z^n = r^n(\cos \theta + i \sin \theta)^n = r^n(\cos n\theta + i \sin n\theta) = R(\cos \alpha + i \sin \alpha).$$

Hence $r^n = R$; that is, $r = \sqrt[n]{R}$ (positive n^{th} root of R) and $n\theta = \alpha + 2k\pi$, $k \in \mathbb{Z}$, that is, $\theta = \frac{\alpha + 2k\pi}{n}$, $k \in \mathbb{Z}$.

$$\therefore \boxed{z = \sqrt[n]{R} \left(\cos \frac{\alpha + 2k\pi}{n} + i \sin \frac{\alpha + 2k\pi}{n} \right)}$$



has n distinct values with $k = 0, 1, 2, \dots, n-1$ and any other value of k will produce a value identical to one of these.

The modulus of all the n^{th} roots of w are identical and is equal to $\sqrt[n]{R}$. Hence these n roots are represented by points on a circle having radius $\sqrt[n]{R}$. Each of the n^{th} roots has a different argument. As k increases, the argument grows by increasing $\frac{2\pi}{n}$.

Example 3.6.4 Find the n^{th} roots of 1 with (i) $n = 3$ (ii) $n = 6$.

SOLUTION.

$$1 = \cos 0 + i \sin 0.$$

(i) The cube roots of unity are given by

$$z_k = \cos \frac{0 + 2k\pi}{3} + i \sin \frac{0 + 2k\pi}{3}, \quad k = 0, 1, 2$$

i.e

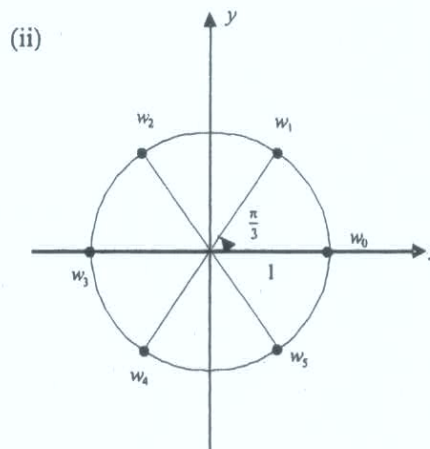
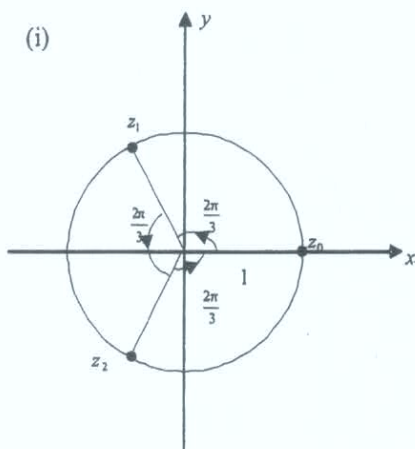
$$z_0 = \cos 0 + i \sin 0 = 1$$

$$z_1 = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$$

$$z_2 = \cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$$

(ii) The 6^{th} roots of unity are given by

$$z_k = \cos \frac{0 + 2k\pi}{6} + i \sin \frac{0 + 2k\pi}{6}, \quad k = 0, 1, 2, 3, 4, 5$$



i.e

$$z_0 = 1$$

$$z_1 = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3} = \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

$$z_2 = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$$

$$z_3 = \cos \frac{3\pi}{3} + i \sin \frac{3\pi}{3} = -1$$

$$z_4 = \cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$$

$$z_5 = \cos \frac{5\pi}{3} + i \sin \frac{5\pi}{3} = \frac{1}{2} - \frac{\sqrt{3}}{2}i$$

Example 3.6.5 Find the two square roots of $w = 2 + 2\sqrt{3}i$. Give your answer in Cartesian form.

SOLUTION.

$|w| = \sqrt{2^2 + (2\sqrt{3})^2} = \sqrt{4 + 12} = 4$. Let $\theta = \text{Arg}(w)$. $(2, 2\sqrt{3})$ is in the first quadrant, $\tan \theta = \frac{2\sqrt{3}}{2} = \sqrt{3}$, $0 \leq \theta \leq \frac{\pi}{2}$. $\therefore \theta = \text{Arg}(w) = \frac{\pi}{3}$ and $w = 4 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$.

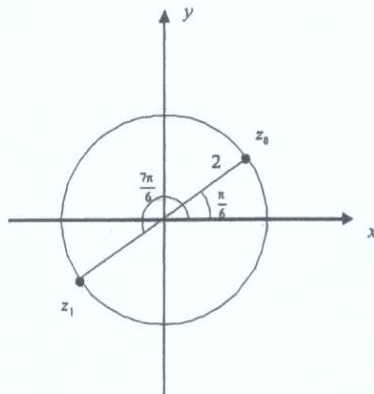
By formula:

$$z_k = \sqrt{4} \left(\cos \frac{\frac{\pi}{3} + 2k\pi}{2} + i \sin \frac{\frac{\pi}{3} + 2k\pi}{2} \right), \quad k = 0, 1$$

are the square roots of w .

$$z_0 = 2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right) = 2 \left(\frac{\sqrt{3}}{2} + \frac{1}{2}i \right) = \sqrt{3} + i$$

$$z_1 = 2 \left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6} \right) = 2 \left(-\frac{\sqrt{3}}{2} - \frac{1}{2}i \right) = -\sqrt{3} - i.$$



EXERCISE.

Use a different method to find a solution of the above example.

Example 3.6.6 Solve the equation $z^4 = -8i$ and indicate your solutions in the Argand plane.

SOLUTION.

$$-8i = 8\left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}\right)$$

$$\begin{aligned} z_k &= \sqrt[4]{8} \left(\cos \frac{\frac{3\pi}{2} + 2k\pi}{4} + i \sin \frac{\frac{3\pi}{2} + 2k\pi}{4} \right), \quad k = 0, 1, 2, 3 \\ &= \sqrt[4]{8} \left(\cos \frac{(4k+3)\pi}{8} + i \sin \frac{(4k+3)\pi}{8} \right), \quad k = 0, 1, 2, 3 \end{aligned}$$

$$z_0 = \sqrt[4]{8} \left(\cos \frac{3\pi}{8} + i \sin \frac{3\pi}{8} \right)$$

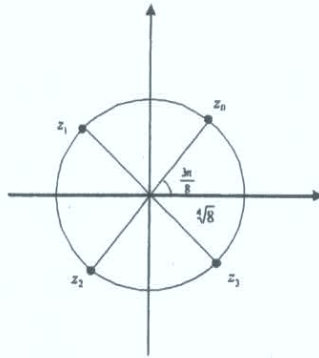
$$z_1 = \sqrt[4]{8} \left(\cos \frac{7\pi}{8} + i \sin \frac{7\pi}{8} \right)$$

$$z_2 = \sqrt[4]{8} \left(\cos \frac{11\pi}{8} + i \sin \frac{11\pi}{8} \right)$$

$$z_3 = \sqrt[4]{8} \left(\cos \frac{15\pi}{8} + i \sin \frac{15\pi}{8} \right)$$

3.7 Complex equations and their Cartesian forms

An equation with complex numbers as variables is called a complex equation. For example,



- (i) $|z| = 2$;
- (ii) $\text{Arg}(z) = \frac{\pi}{3}$;
- (iii) $\left| \frac{z+2i}{z+3-i} \right| = 2$;
- (iv) $|z - 4i| + |z + 3| = 2$ where $z \in \mathbb{C}$.

are complex equations.

Letting $z = x + iy$, complex equations can be written in Cartesian form, i.e. in terms of x and y .

The set of all $z = x + iy$ satisfying a complex equation forms a curve in the xy -plane. This curve is called the locus of z .

Example 3.7.1 What is the locus of z satisfying $|z| = 2$?

SOLUTION.

Let $z = x + yi$. Then $|z| = 2$ becomes $\sqrt{x^2 + y^2} = 2$ i.e. $x^2 + y^2 = 4$.
 $\therefore$ the locus of z is the circle with center $(0, 0)$ and radius 2.

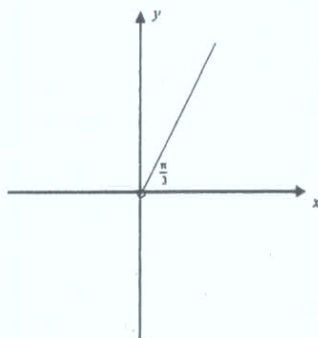
Example 3.7.2 What is the Cartesian form of the equation $\text{Arg}(z) = \frac{\pi}{3}$ and what curve is represented by this equation?

SOLUTION.

Let $z = x + yi$, $x, y \in \mathbb{R}$. Then

$$\begin{aligned} \operatorname{Arg}(z) = \frac{\pi}{3} &\Leftrightarrow x \geq 0, y \geq 0 \text{ and } \tan \frac{\pi}{3} = \frac{y}{x} = \sqrt{3} \\ &\Leftrightarrow y = \sqrt{3}x, x > 0 \end{aligned}$$

The required locus is a straight line consisting of points $(x, \sqrt{3}x)$ with $x > 0$.



Example 3.7.3 Describe the locus represented by

$$\left| \frac{z + 2i}{z - i} \right| = 2 \quad (3.3)$$

SOLUTION.

The equation (3.3) can be written as $|z + 2i| = 2|z - i|$ or

$$|z + 2i|^2 = 4|z - i|^2 \quad (3.4)$$

Let $z = x + yi$. Then equation (3.4) becomes

$$\begin{aligned} |(x + yi) + 2i|^2 &= 4|(x + yi) - i|^2 \\ |x + (y + 2)i|^2 &= 4|x + (y - 1)i|^2 \\ x^2 + (y + 2)^2 &= 4(x^2 + (y - 1)^2) \\ x^2 + y^2 + 4y + 4 &= 4(x^2 + y^2 - 2y + 1) \\ 3x^2 + 3y^2 - 12y &= 0 \\ x^2 + y^2 - 4y &= 0 \\ x^2 + (y - 2)^2 &= 4 \end{aligned}$$

The locus is a circle with center $(0, 2)$ and radius 2.

Example 3.7.4 If $|z| = 3$, what is the locus of the complex number $2z+1$?

SOLUTION.

Let $z_1 = 2z + 1$. Then $z = \frac{1}{2}(z_1 - 1)$.

$$|z| = 3 \Leftrightarrow \left| \frac{1}{2}(z_1 - 1) \right| = 3 \Leftrightarrow |z_1 - 1| = 6$$

Let $z_1 = x + yi$. Then

$$|(x + yi) - 1| = 6$$

$$|(x - 1) + yi| = 6$$

$$(x - 1)^2 + y^2 = 36$$

This is the equation of the circle with center $(1, 0)$ and radius 6.

Chapter 4

Polynomials

4.1 Introduction

A polynomial is an expression of the form

$$a_0 + a_1x + \dots + a_nx^n \quad (4.1)$$

where n is a non-negative integer and $a_0, a_1, \dots, a_n$ are constants. It is customary to call x an indeterminate. For $0 \leq i \leq n$, a_i is called the coefficient of x^i in the polynomial. The largest integer m for which $a_m \neq 0$ is called the degree of the polynomial (4.1); a_m is called the leading coefficient. If all the coefficients are complex numbers, then the polynomial (4.1) is called a complex polynomial. If all the coefficients are real numbers, then the polynomial (4.1) is called a real polynomial.

Example 4.1.1

(i)	-3	degree 0	real polynomial
(ii)	$8 + 6x$	degree 1	real polynomial
(iii)	$-9 + 2x + \sqrt{3}x^2$	degree 2	real polynomial
(iv)	$1 + (-2)x + 0x^2 + 3x^3 + 4x^4$	degree 4	real polynomial
(v)	$4 + ix + 5x^2 + 7x^3$	degree 3	complex polynomial

Remark 4.1.1 (i) *Polynomials of degree 1 are called linear.*

(ii) *Polynomials of degree 2 are called quadratic.*

(iii) *Polynomials of degree 3 are called cubic.*

Two polynomials $f(x)$ and $g(x)$ are called equal if their expression are identical except for the terms with zero coefficients. We therefore consider a polynomial as being unchanged by the insertion or omission, of any number of terms with zero coefficients. In particular, we may write

$$(i) \quad 0 + 0x + 4x^2 = 4x^2$$

$$(ii) \quad 1 + 2x + (-7)x^2 + 0x^3 = 1 + 2x + (-7)x^2$$

$$(iii) \quad a_0 + 0x + 0x^2 + 0x^3 = a_0$$

With this understanding of zero coefficients, if $f(x)$ is a polynomial and i is an arbitrary positive integer, we may speak of the coefficient of x^i in $f(x)$. For example, in the polynomial $1 + 2x + 4x^2$, the coefficient of x^5 is zero. Zero polynomial is the polynomial with all coefficients equal to zero. (Note that the degree of the zero polynomial is undefined.) The zero polynomial is denoted by 0.

Example 4.1.2 If $ax^2 + bx + c = 2x^2 + 3x - 5$, then $a = 2, b = 3, c = -5$.
If $ax^2 = bx + c = 0$, then $a = b = c = 0$.

4.2 Operations on polynomials

4.2.1 Addition and subtraction

Consider two polynomials

$$f(x) = a_0 + a_1x + \dots + a_mx^m,$$

$$g(x) = b_0 + b_1x + \dots + b_nx^n.$$

Assume that $m \geq n$. We define

$$\begin{aligned} f(x) + g(x) &= (a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2 + \dots + (a_n + b_n)x^n \\ &\quad + a_{n+1}x^{n+1} + \dots + a_mx^m. \end{aligned}$$

In other words, for every non-negative integer i the coefficient of x^i in $f(x) + g(x)$ is the sum of the coefficients of x^i in $f(x)$ and in $g(x)$.

Example 4.2.1 If $f(x) = 2x + x^2$, $g(x) = 2 + 3x + x^3$, then

$$f(x) + g(x) = (0 + 2) + (2 + 3)x + (1 + 0)x^2 + (0 + 1)x^3 = 2 + 5x + x^2 + x^3.$$

Similarly, we can define $f(x) - g(x)$. The coefficient of x^i in $f(x) - g(x)$ is $a_i - b_i$ if a_i is the coefficient of x^i in $f(x)$ and b_i is the coefficient of x^i in $g(x)$ where i is any non-negative integer.

Example 4.2.2 If $f(x) = 2x + x^2$ and $g(x) = 2 + 3x + x^3$, then

$$f(x) - g(x) = (0 - 2) + (2 - 3)x + (1 - 0)x^2 + (0 - 1)x^3 = -2 - x + x^2 - x^3$$

(Note that we have used the convention that $-cx^2 = +(-c)x^2$ in a polynomial.)

Remark 4.2.1 For any two polynomials $f(x), g(x)$, we have $f(x) + g(x) = g(x) + f(x)$.

4.2.2 Multiplication

Let

$$f(x) = a_0 + a_1x + \dots + a_mx^m$$

$$g(x) = b_0 + b_1x + \dots + b_nx^n$$

be two polynomials. We define

$$f(x)g(x) = a_0b_0 + (a_0b_1 + a_1b_0)x + (a_0b_2 + a_1b_1 + a_2b_0)x^2 + \dots + (a_mb_n)x^{m+n}.$$

More explicitly, if $i \leq m + n$ is an arbitrary non-negative integer, the coefficient of x^i in the product is

$$a_0b_i + a_1b_{i-1} + \dots + a_ib_0$$

with the understanding that $a_k = 0$ if $k > m$ and $b_k = 0$ if $k > n$.

Note that the following distributive laws holds:

$$f(x)(g(x) + h(x)) = f(x)g(x) + f(x)h(x)$$

$$(g(x) + h(x))f(x) = g(x)f(x) + h(x)f(x)$$

for any three polynomials $f(x), g(x)$ and $h(x)$.

Example 4.2.3

$$\begin{aligned} & (2 + 3x + x^3)(2x + x^2) \\ &= 4x + (3 \times 2 + 2 \times 1)x^2 + (3 \times 1)x^3 + (1 \times 2)x^4 + (1 \times 1)x^5 \\ &= 4x + 8x^2 + 3x^3 + 2x^4 + x^5 \end{aligned}$$

(Note that we can compute the above multiplication following the distributive law as follows:

$$\begin{aligned}
 & (2 + 3x + x^3)(2x + x^2) \\
 = & (2 + 3x + x^3)(2x) + (2 + 3x + x^3)(x^2) \\
 = & 4x + 6x^2 + 2x^4 + 2x^2 + 3x^3 + x^5 \\
 = & 4x + 8x^2 + 3x^3 + 2x^4 + x^5.)
 \end{aligned}$$

Remark 4.2.2 If $f(x)$ and $g(x)$ are nonzero polynomials, then

$$\deg(f(x) + g(x)) = \deg(f(x)) + \deg(g(x)).$$

4.2.3 Substitution

Let $P(x) = a_0 + a_1x + \dots + a_nx^n$ be a polynomial and c a fixed number. Then we define

$$P(c) = a_0 + a_1c + \dots + a_nc^n.$$

If $P(x) = f(x) + g(x)$, $q(x) = f(x)g(x)$, then

$$P(c) = f(c) + g(c), \quad q(c) = f(c)g(c).$$

Example 4.2.4 If $P(x) = (x - a)d(x) + b$, a, b constants, then $P(a) = (a - a)d(a) + b = b$.

4.2.4 Division Algorithm

Let $f(x)$ and $g(x)$ be two polynomials with $g(x) \neq 0$. Then there exist unique polynomials $q(x)$ and $r(x)$ such that

$$f(x) = q(x)g(x) + r(x) \quad \text{where } r(x) = 0 \text{ or } \deg r(x) < \deg g(x). \quad (4.2)$$

(Here, $q(x)$ is called the quotient and $r(x)$ is called the remainder in the division of $f(x)$ by $g(x)$.)

Remark 4.2.3 (i) If $f(x) = 0$ or $\deg f(x) < \deg g(x)$, then (4.2) is satisfied with $q(x) = 0$ and $r(x) = f(x)$.

(ii) If $\deg f(x) \geq \deg g(x)$, then we have $\deg f(x) = \deg g(x) + \deg q(x)$

4.3 Remainder Theorem

Theorem 4.3.1 (*Remainder Theorem*) If $f(x)$ is a polynomial and c is a constant, then the remainder in the division of $f(x)$ by $x - c$ is $f(c)$.

PROOF.

By the division algorithm, there exist polynomials $q(x)$ and $r(x)$ such that

$$f(x) = (x - c)q(x) + r(x) \quad (4.3)$$

where $r(x) = 0$ or $\deg r(x) < \deg(x - c)$. Since degree of $x - c$ is 1, it follows that $r(x) = 0$ or $\deg r(x) = 0$. This implies that $r(x)$ is a constant, i.e. $r(x) = b$ for some constant b . By (4.3), we have $f(c) = (c - c)q(c) + b = b$. $\therefore f(c)$ is the remainder.

Remark 4.3.1 When $f(x)$ is divided by $ax - b$ or $x - \frac{b}{a}$, where $a \neq 0$, the remainders are the same (however the quotients are different).

PROOF.

Note that $f(x) = (x - \frac{b}{a})q(x) + r$ for some polynomial $q(x)$ and some constant r . Then $f(x) = (ax - b)(\frac{1}{a}q(x)) + r$. Hence the remainders are the same.

Example 4.3.1 What is the remainder when $x^3 + (2 + 3i)x^2 - x - i$ is divided by $2x + 1$?

SOLUTION.

Let $f(x) = x^3 + (2 + 3i)x^2 - x - i$. Since $2x + 1 = 2(x - (-\frac{1}{2}))$, it follows from the previous remark that the remainder is $f(-\frac{1}{2})$.

$$\therefore \text{remainder} = -\frac{1}{8} + \frac{1}{4}(2 + 3i) + \frac{1}{2} - i = \frac{1}{8}(7 - 2i)$$

Example 4.3.2 Find the remainder when $f(x) = 3x^4 + 12x^2 - 5x + 36$ is divided by $x^2 + 2x - 3$.

SOLUTION.

Since $x^2 + 2x - 3 = (x - 1)(x + 3)$ is of degree 2, it follows that

$$f(x) = (x - 1)(x + 3)q(x) + ax + b$$

for some constants a, b and some polynomial $q(x)$.

$$f(1) = a + b = 3 + 12 - 5 + 36 = 46$$

$$f(-3) = -3a + b = 243 + 108 + 15 + 36 = 402$$

$$4a = -356, \quad \therefore a = -89$$

$$\therefore b = 46 - a = 46 + 89 = 135$$

$\therefore$ remainder $= -89x + 135$.

Remark 4.3.2 The above method cannot be used for finding quotients. The following methods (long divisions and synthetic divisions) allow us to find both the remainder and the quotient.

4.3.1 Long division

For each of the following polynomials $P(x)$ and $d(x)$, find the quotient and the remainder when $P(x)$ is divided by $d(x)$:

(a) $P(x) = 1 + 2x^3 - 3x^2, d(x) = x - 2$;

(b) $P(x) = x + 2x^4 - 3x^3 + ix^2 - 2, d(x) = x^2 - 3x + 2$

SOLUTION.

(a)

$$\begin{array}{r}
 \overline{2x^2 + x + 2} \\
 x-2 \) \ 2x^3 - 3x^2 + 0x + 1 \\
 \underline{2x^3 - 4x^2} \\
 x^2 + 0x + 1 \\
 \underline{x^2 - 2x} \\
 2x + 1 \\
 \underline{2x - 4} \\
 5
 \end{array}$$

$\therefore p(x) = (x-2)(2x^2 + x + 2) + 5$
 quotient = $2x^2 + x + 2$, remainder = 5.

(b)

$$\begin{array}{r}
 \begin{array}{r}
 x^2 - 3x + 2 \quad) \quad \begin{array}{r}
 2x^4 - 3x^3 + ix^2 + x - 2 \\
 \underline{2x^4 - 6x^3 + 4x^2} \\
 3x^3 + (-4+i)x^2 + x - 2 \\
 \underline{3x^3 - 9x^2 + 6x} \\
 (5+i)x^2 - 5x - 2 \\
 \underline{(5+i)x^2 - (15+3i)x + (10+2i)} \\
 (10+3i)x - 12 - 2i
 \end{array}
 \end{array}
 \end{array}$$

$\therefore p(x) = (x^2 - 3x + 2)(2x^2 + 3x + (5+i)) + ((10+3i)x - 12 - 2i)$
 quotient = $2x^2 + 3x + (5+i)$, remainder = $(10+3i)x - 12 - 2i$.

4.3.2 Synthetic divisions

Consider the polynomial

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \dots + a_1 x + a_0 \quad (a_n \neq 0)$$

of degree n and the divisor $x - b$. We wish to find the quotient $q(x)$ and the remainder r such that

$$P(x) = (x - b)q(x) + r. \quad (4.4)$$

(r is a constant, $\deg q(x) = n - 1$). Let

$$q(x) = c_n x^{n-1} + c_{n-1} x^{n-2} + \dots + c_2 x + c_1 \quad (4.5)$$

and

$$r = c_0 \quad (4.6)$$

We wish to find c_i ($i = 0, 1, 2, \dots, n$).

By (4.4), (4.5) and (4.6), we have

$$\begin{aligned}
 & a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \dots + a_1 x + a_0 \\
 = & (x - b)(c_n x^{n-1} + c_{n-1} x^{n-2} + \dots + c_2 x + c_1) + c_0
 \end{aligned}$$

By comparing the coefficients, we obtain

$$\left. \begin{array}{lcl} x^n & : & a_n = c_n \\ x^{n-1} & : & a_{n-1} = -bc_n + c_{n-1} \\ x^{n-2} & : & a_{n-2} = -bc_{n-1} + c_{n-2} \\ & \dots & \\ x & : & a_1 = -bc_2 + c_1 \\ x^0 & : & a_0 = -bc_1 + c_0 \end{array} \right\} \text{i.e.} \quad \begin{array}{lcl} c_n & = & a_n \\ c_{n-1} & = & a_{n-1} + bc_n \\ c_{n-2} & = & a_{n-2} + bc_{n-1} \\ & \dots & \\ c_1 & = & a_1 + bc_2 \\ c_0 & = & a_0 + bc_1 \end{array}$$

From the above table, we can obtain c_i by constructing the following table:

$$\begin{array}{ccccccc} & x^n & x^{n-1} & x^{n-2} & \dots & x^2 & x & x^0 = 1 \\ \hline \begin{array}{c} b \\ + \end{array} & \begin{array}{c} a_n \\ \downarrow \\ c_n \end{array} & a_{n-1} & a_{n-2} & \dots & a_2 & a_1 & a_0 \\ & & \nearrow bc_n & \nearrow bc_{n-1} & \dots & \nearrow bc_3 & \nearrow bc_2 & \nearrow bc_1 \\ a_n = & \hline & c_{n-1} & c_{n-2} & \dots & c_2 & c_1 & c_0 \end{array}$$

Example 4.3.3 For each pair of polynomials $P(x)$ and $d(x)$, find the quotient and the remainder when $P(x)$ is divided by $d(x)$.

- (a) $P(x) = x^4 - 2x^3 + 4x - 1$, $d(x) = x - 3$.
- (b) $P(x) = 2x^3 + 3x^2 - (5 + 2i)x - 9i$, $d(x) = x + (2 - i)$
- (c) $P(x) = 10x^4 + 7x^3 + x^2$, $d(x) = 2x + 1$
- (d) $P(x) = 3x^4 + 12x^2 - 5x + 36$, $d(x) = x^2 + 2x - 3$

SOLUTION.

- (a) divisor = $x - 3$
quotient, $q(x) = x^3 + x^2 + 3x + 13$; remainder, $r(x) = 38$
- (b) divisor = $x + (2 - i)$
quotient, $q(x) = 2x^2 + (-1 + 2i)x - (5 + 7i)$; remainder, $r(x) = 17$

$$\begin{array}{r|rrrrr}
 & x^4 & x^3 & x^2 & x & 1 \\
 3 & 1 & -2 & 0 & 4 & -1 \\
 + & & 3 & 3 & 9 & 39 \\
 \hline
 & 1 & 1 & 3 & 13 & 38
 \end{array}$$

$$\begin{array}{r|rrrr}
 & x^3 & x^2 & x & 1 \\
 -2+i & 2 & 3 & -5-2i & -9i \\
 + & & -4+2i & -5i & 17+9i \\
 \hline
 & 2 & -1+2i & -5-7i & 17
 \end{array}$$

(c) divisor = $2x + 1$

$$\begin{array}{r|rrrrr}
 & x^4 & x^3 & x^2 & x & 1 \\
 -\frac{1}{2} & 10 & 7 & 1 & 0 & 0 \\
 + & & -5 & -1 & 0 & 0 \\
 \hline
 & 10 & 2 & 0 & 0 & 0
 \end{array}$$

quotient, $q(x) = 5x^3 + x^2$; remainder, $r(x) = 0$

(d) divisor = $x^2 + 2x - 3$

From the first division,

$$p(x) = (3x^3 + 3x^2 + 15x + 10)(x - 1) + 46.$$

From the second division,

$$3x^3 + 3x^2 + 15x + 10 = (3x^2 - 6x + 33)(x + 3) - 89$$

$$\begin{aligned}
 \therefore p(x) &= [(3x^2 - 6x + 33)(x + 3) - 89](x - 1) + 46 \\
 &= (3x^2 - 6x + 33)(x + 3)(x - 1) - 89(x - 1) + 46 \\
 &= (3x^2 - 6x + 33)(x^2 + 2x - 3) - 89x + 135
 \end{aligned}$$

quotient, $q(x) = 3x^2 - 6x + 33$; remainder, $r(x) = -89x + 135$.

	x^4	x^3	x^2	x	1
1	3	0	12	-5	36
+		3	3	15	10
-3	3	3	15	10	46
+		-9	18	-99	
	3	-6	33	-89	

A polynomial $d(x)$ is called a factor of another polynomial $f(x)$ if there exists a polynomial $h(x)$ such that

$$f(x) = d(x)h(x)$$

If $d(x)$ is a factor of $f(x)$, we say also that $f(x)$ is divisible by $d(x)$ or that $f(x)$ is a multiple of $d(x)$.

Remark 4.3.3 $d(x)$ is a factor of $f(x)$ if and only if the remainder is zero when $f(x)$ is divided by $d(x)$.

4.4 Factor Theorem

Theorem 4.4.1 The polynomial $x - c$ is a factor of $f(x)$ if and only if $f(c) = 0$.

PROOF.

The polynomial $x - c$ is a factor of $f(x)$ if and only if the remainder in the division of $f(x)$ by $x - c$ is zero. Since the remainder is equal to $f(c)$, it follows that $x - c$ is a factor of $f(x)$ if and only if $f(c) = 0$.

Example 4.4.1 Determine whether or not $x - a$ is a factor of $f(x) = x^3 - 2x^2 + x - 2$ when $a = 1, i$, or $-i$.

SOLUTION.

$$\begin{aligned} f(1) &= -2 \neq 0 & \therefore x - 1 \text{ is not a factor of } f(x) \\ f(i) &= -i + 2 + i - 2 = 0 & \therefore x - i \text{ is a factor of } f(x) \\ f(-i) &= i + 2 - i - 2 = 0 & \therefore x + i \text{ is a factor of } f(x) \end{aligned}$$

Definition 4.4.1 If $P(x)$ is a polynomial and $P(a) = 0$ ($a \in \mathbb{C}$), then a is called a zero of $P(x)$ or a root of the equation $P(x) = 0$. (A zero of $P(x)$ is sometimes called a root of $P(x)$.)

Remark 4.4.1 The following statements are equivalent:

- (i) $P(a) = 0$;
- (ii) $x - a$ is a factor of $P(x)$;
- (iii) a is a zero of $P(x)$;
- (iv) a is a root of the equation $P(x) = 0$.

Example 4.4.2 $P(x) = x^3 - 2x^2 + x - 2 = (x - i)(x + i)(x - 2)$.

$\therefore P(i) = P(-i) = P(2) = 0$;

$\therefore x - i, x + i, x - 2$ are all factors of $P(x)$;

$i, -i, 2$ are roots of $P(x)$;

$i, -i, 2$ are roots of the polynomial equation $P(x) = 0$.

4.5 Fundamental Theorem of Algebra

The following important theorem was first proved by the famous German mathematician Carl Friedrich Gauss (1777 – 1855).

Theorem 4.5.1 Every polynomial of degree $n > 0$ has exactly n roots in $\mathbb{C}$, counting multiplicities.

This theorem says that every polynomial $P(x)$ with degree $n > 0$ can be factorized as follows:

$$P(x) = c(x - \alpha_1)(x - \alpha_2) \dots (x - \alpha_n)$$

for some complex numbers $c, \alpha_1, \alpha_2, \dots, \alpha_n$.

Example 4.5.1 Let $P(x) = 2x^3 - 3x^2 + kx + 3$ (k is a constant). Find the value of k if $-\frac{1}{2}$ is a root of $P(x)$ and then obtain all the other roots of $P(x)$.

$$\begin{array}{r|rrrr}
 & x^3 & x^2 & x & 1 \\
 -\frac{1}{2} & 2 & -3 & 4 & 3 \\
 \hline
 + & & -1 & 2 & -3 \\
 \hline
 & 2 & -4 & 6 & 0
 \end{array}$$

SOLUTION.

$-\frac{1}{2}$ is a root (zero) of $P(x)$.

$$\Rightarrow P(-\frac{1}{2}) = -\frac{1}{4} - \frac{3}{4} - \frac{k}{2} + 3 = 2 - \frac{k}{2} = 0 \Rightarrow k = 4.$$

$$\therefore P(x) = (x + \frac{1}{2})(2x^2 - 4x + 6) = (2x + 1)(x^2 - 2x + 3)$$

The solution of $x^2 - 2x + 3 = 0$ is given by

$$x = \frac{2 \pm \sqrt{4 - 12}}{2} = \frac{2 \pm 2\sqrt{2}i}{2} = 1 \pm \sqrt{2}i$$

$\therefore$ the roots of $P(x)$ are $-\frac{1}{2}, 1 + \sqrt{2}i, 1 - \sqrt{2}i$

Example 4.5.2 Find the polynomial $P(x)$ of degree 3 such that $P(1) = P(2) = 0$, $P(3) = 130$ and $P(-3) = -20$.

SOLUTION.

Since $P(1) = P(2) = 0$, it follows that $(x - 1)(x + 2)$ is a factor of $P(x)$ and hence $P(x) = (x - 1)(x + 2)(ax + b)$ for some constants a, b .

$$\left. \begin{array}{l} P(3) = 2 \times 5 \times (3a + b) = 130 \\ P(-3) = -4 \times (-1) \times (-3a + b) = -20 \end{array} \right\} \text{i.e. } \begin{array}{l} 3a + b = 13 \\ -3a + b = -5 \end{array}$$

These give $b = 4$ and $a = 3$.

$$\therefore P(x) = (x - 1)(x + 2)(3x + 4) = 3x^3 + 7x^2 - 2x - 8.$$

Definition 4.5.1 Let k be a positive integer. A number a is said to be a root of multiplicity k of $P(x)$ if

(i) $(x - a)^k$ is a factor of $P(x)$.

(ii) $(x - a)^{k+1}$ is not a factor of $P(x)$.

A root of multiplicity two is called a double root.

Example 4.5.3 Suppose that $f(x) = (x - 1)^3(x + 2)(x - i)^2$. Then 1 is a root of multiplicity 3; -2 is a root of multiplicity 1; i is a root of multiplicity 2.

Definition 4.5.2 Let c be a root of multiplicity k of a polynomial $P(x)$.

- (i) If $k = 1$, then c is called a simple root.
- (ii) If $k > 1$, then c is called a multiple root.

Let $f(x) = a_0 + a_1x + \dots + a_nx^n$ be a polynomial. Then the polynomial $a_1 + 2a_2x + \dots + na_nx^{n-1}$ is called the derivative of $f(x)$ and is denoted by $f'(x)$. The derivative of $f'(x)$ is called the second derivative of $f(x)$ and is denoted by $f''(x)$. Continuing this process, we can define the k^{th} derivative of a polynomial $f(x)$ for $k > 2$. This k^{th} derivative of $f(x)$ is denoted $f^{(k)}(x)$.

Example 4.5.4 Let $f(x) = 3 + 4x + 5x^2 + 6x^4$. Then $f'(x) = 4 + 10x + 24x^3$.

The derivative has the following properties: For polynomials $f(x), g(x)$,

- (i) If $p(x) = f(x) + g(x)$, then $p'(x) = f'(x) + g'(x)$;
- (ii) If $p(x) = f(x)g(x)$, then $p'(x) = f'(x)g(x) + f(x)g'(x)$;
- (iii) If $p(x) = cf(x)$ (c a constant), then $p'(x) = cf'(x)$;

Using the above properties, one can show that

- (iv) If $p(x) = c(x - a)^m, m \geq 1$, then $p'(x) = cm(x - a)^{m-1}$.

Suppose that a is root of multiplicity 2 of a polynomial $f(x)$. Then $f(x) = (x - a)^2q(x)$ for some polynomial $q(x)$ such that $x - a$ is not a factor of $q(x)$. Then

$$f'(x) = 2(x - a)q(x) + (x - a)^2q'(x)$$

and hence $f'(a) = 0$. Since

$$f''(x) = (2q(x) + 2(x - a)q'(x)) + (2(x - a)q'(x) + (x - a)^2q''(x)),$$

we have $f''(a) = 2q(a) \neq 0$ since $x - a$ is not a factor of $q(x)$.

4.6 Multiple Root Theorem

Let k be a positive integer. Then a polynomial $P(x)$ has a root of multiplicity k if and only if

$$P(a) = P'(a) = P''(a) = \dots = P^{(k-1)}(a) = 0 \text{ and } P^{(k)}(a) \neq 0$$

Example 4.6.1 Find all the roots (zeros) of $P(x) = x^4 - 6x^2 + 8x - 3$ if one of them is of multiplicity three.

SOLUTION.

$$P(x) = x^4 - 6x^2 + 8x - 3,$$

$$P'(x) = 4x^3 - 12x + 8,$$

$$P''(x) = 12x^2 - 12.$$

Let a be the root of multiplicity 3. Then $P''(a) = 12a^2 - 12 = 0$. $\therefore a = \pm 1$. Since $P(-1) = 1 - 6 - 8 - 3 \neq 0$, 1 is the root of multiplicity 3. Hence $P(x) = (x - 1)^3(cx + d)$ for some constants c and d . By comparing the coefficients, we have $c = 1, d = 3$.

$$\therefore P(x) = (x - 1)^3(x + 3)$$

$$\therefore P(x) \text{ has four roots } 1, 1, 1, -3$$

Recall that a real polynomial is a polynomial such that all coefficients are real numbers.

Theorem 4.6.1 If $z = a + bi$ ($a, b \in \mathbb{R}, b \neq 0$) is a root of the real polynomial $P(x)$, then its conjugate $a - bi$ is also a root of $P(x)$.

PROOF.

Let $N(x) = (x - z)(x - \bar{z}) = [x - (a + bi)][x - (a - bi)] = x^2 - 2ax + a^2 + b^2$. Thus $N(x)$ is a real polynomial. Hence when $P(x)$ is divided by the real polynomial $N(x)$ of degree 2, the remainder is a real polynomial of the form $cx + d$, i.e.

$$P(x) = (x - z)(x - \bar{z})q(x) + cx + d.$$

Since z is a root of $P(x)$, it follows that

$$P(z) = c(a + bi) + d = 0.$$

Hence

$$ca + d = 0 \quad (4.7)$$

$$cb = 0 \quad (4.8)$$

Since $b \neq 0$, we have $c = 0$ (by (4.8)) and hence by (4.7), $d = 0$.

$\therefore P(x) = (x - z)(x - \bar{z})q(x)$.

This shows that $\bar{z}$ is a root of $P(x)$.

Example 4.6.2 The polynomial $P(x) = (x - i)(x + 2)$ is not a real polynomial. It has a complex root i . However the conjugate $-i$ is not a root of $P(x)$.

Example 4.6.3 Find all roots of the equation $2x^3 - 7x^2 + 10x - 6 = 0$ if one of its roots is $1 + i$.

SOLUTION.

Since $1 + i$ is a root, it follows that $1 - i$ is also a root of the equation.

	x^3	x^2	x	1
$1+i$	2	-7	10	-6
	+	$2+2i$	$-7-3i$	6
$1-i$	2	$-5+2i$	$3-3i$	0
	+	$2-2i$	$-3+3i$	
	2	-3		0

$$\therefore P(x) = [x - (1 + i)][x - (1 - i)][2x - 3]$$

$\therefore$ it has 3 roots $1 + i, 1 - i, \frac{3}{2}$.

Definition 4.6.1 A rational number $\frac{p}{q}$ (p, q integers, $q \neq 0$) is said to be in lowest terms if p, q are relatively prime i.e. p, q have no common factors except ± 1 .

Example 4.6.4 The rational number $\frac{3}{4}$ is in lowest terms; $\frac{6}{8}$ is not in lowest terms.

Theorem 4.6.2 (Rational roots) Let $f(x) = a_0 + a_1x + \dots + a_{n-1}x^{n-1} + a_nx^n$ be a polynomial with coefficients which are integers. If the rational number $\frac{p}{q}$ is a root of $f(x) = 0$ in lowest term. Then $p \mid a_0$ and $q \mid a_n$, i.e.

$$a_0 = ps \text{ and } a_n = qt \text{ for some integers } s, t$$

($c \mid d$ (c, d integers) denotes that c divides d .)

PROOF.

$\frac{p}{q}$ is a root of $f(x) = 0$

$$\Rightarrow a_n \left(\frac{p}{q}\right)^n + a_{n-1} \left(\frac{p}{q}\right)^{n-1} + \dots + a_1 \left(\frac{p}{q}\right) + a_0 = 0.$$

Multiplying both sides of this equation by q^n , we obtain

$$a_n p^n + a_{n-1} p^{n-1} q + \dots + a_1 p q^{n-1} + a_0 q^n = 0 \quad (4.9)$$

(4.9) can also be written as

$$p(a_n p^{n-1} + a_{n-1} p^{n-2} q + \dots + a_1 q^{n-1}) = -a_0 q^n.$$

$$\therefore p \mid a_0 q^n.$$

Since $p \mid a_0 q^n$, p and q are relatively prime, it follows that $p \mid a_0$. Note that (4.9) can be written as

$$q(a_{n-1} p^{n-1} + \dots + a_1 p q^{n-2} + a_0 q^{n-1}) = -a_n p^n.$$

$\therefore q \mid a_n p^n$. Since $q \mid a_n p^n$, p and q are relatively prime, it follows that $q \mid a_n$.

Example 4.6.5 Solve the equation $x^4 - x^3 - 3x^2 + 4x - 4 = 0$.

SOLUTION.

Let $f(x) = x^4 - x^3 - 3x^2 + 4x - 4 = 0$. If a rational number $\frac{p}{q}$ (in lowest term) is a root of $f(x) = 0$, then $p \mid -4$ and $q \mid 1$ and hence

$$p = \pm 1, \pm 2, \pm 4 \text{ and } q = \pm 1.$$

$\therefore$ the possible rational roots are $\pm 1, \pm 2, \pm 4$.

$$f(1) = -3 \neq 0, f(2) = 0, f(-2) = 0.$$

Hence $f(x) = (x-2)(x+2)q(x)$ for some polynomial $q(x)$

$$\begin{array}{r} x^2 - 4 \quad) \quad \begin{array}{rrrrr} & x^2 & -x & +1 & \\ \hline x^4 & -x^3 & -3x^2 & +4x & -4 \\ \hline & x^4 & & -4x^2 & \\ \hline & & -x^3 & +x^2 & +4x & -4 \\ & & \hline & & -x^3 & & +4x \\ & & & \hline & & & x^2 & & -4 \\ & & & \hline & & & x^2 & & -4 \\ & & & \hline & & & & & 0 \end{array} \end{array}$$

$\therefore f(x) = (x-2)(x+2)(x^2 - x + 1)$. The roots of $x^2 - x + 1 = 0$ are $\frac{1 \pm \sqrt{1-4}}{2} = \frac{1 \pm \sqrt{3}i}{2}$.

Hence $f(x) = 0$ has 4 roots, namely, $2, -2, \frac{1+\sqrt{3}i}{2}, \frac{1-\sqrt{3}i}{2}$.

Example 4.6.6 Show that $3x^2 - x + 5 = 0$ has no rational roots.

SOLUTION.

Let $P(x) = 3x^2 - x + 5$. If $P(x)$ has a rational root $\frac{p}{q}$ in lowest term, then $p \mid 5$ and $q \mid 3$ and hence $p = \pm 1, \pm 5, q = \pm 1, \pm 3$.

$\therefore$ the possible rational roots are $\pm 1, \pm 5, \pm \frac{1}{3}, \pm \frac{5}{3}$.

We check that $P(1) \neq 0, P(-1) \neq 0, P(5) \neq 0, P(-5) \neq 0, P(\frac{1}{3}) \neq 0, P(-\frac{1}{3}) \neq 0, P(\frac{5}{3}) \neq 0, P(-\frac{5}{3}) \neq 0$. This shows that the equation $P(x) = 0$ has no rational roots.